

KairoX ERP — Phase 2

API Reference

Every endpoint of the procurement engine, with purpose, roles, request, response and errors — enough to build the entire frontend against mocks, and enough for a backend developer to use as the contract of record.

Covers: Procurement (Stage 1) · BOM (Stage 2/3) · Inventory (Stage 4) · Supplier PO (Stage 5) · Intelligence (Chat) · Auth

Companion document: *KairoX Phase-2 System & Business Guide* — the business logic, module architecture and screen design behind these endpoints.

How to use this document

Frontend developers. Every endpoint below carries a complete, realistic mock response. Copy them into a mock server (MSW, json-server, or a plain fixtures file) and build the entire UI without a backend. \$29 is a ready-made fixture pack with consistent IDs across every endpoint, so one mocked order flows end to end. \$30 gives the exact call sequences for each user journey.

Backend developers. This is the contract of record: exact paths, exact role gates, exact status codes, exact response keys. Each section names the router file and the service method behind it.

Everyone. Endpoints are grouped by stage, in the order a real order travels through them.

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FOUNDATION

1. Base URL and versioning

```
https://<host>/api/v1
```

Every endpoint in this document is prefixed with `/api/v1`. Paths below are written **relative to that prefix** — so `/procurement/submissions` means `https://<host>/api/v1/procurement/submissions`.

Four route groups appear here:

Group	Prefix	Owned by
Authentication	<code>/auth</code>	<code>users</code> module
The procurement engine	<code>/procurement</code>	Four modules share this prefix by design — Stage 1, 2/3, 4 and 5
Chat	<code>/chat</code>	<code>intelligence</code> module
Health	(no prefix — <code>/health</code> , <code>/ready</code>)	<code>main.py</code>

Why four modules share `/procurement`. The backend was split out of a former monolith, and the URLs were deliberately preserved so no frontend had to change. Do not read anything into the prefix — `/procurement/boms/...` is the BOM module, `/procurement/inventory/...` is the inventory module.

Interactive docs live at `/docs` (Swagger) and `/redoc`.

2. Authentication

Self-issued JWT bearer tokens.

POST `/auth/login`

Request

```
{ "phone": "9876543210", "password": "....." }
```

Response `200`

```
{
  "access_token": "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9...",
  "token_type": "bearer"
}
```

Send it on every subsequent request:

```
Authorization: Bearer <access_token>
```

GET `/auth/me`

Returns the current user — use it on app boot to restore the session and to decide which actions to render.

Response 200

```
{
  "id": "9e1c4a70-0b2d-4c8e-9f11-6a2b3c4d5e6f",
  "name": "Tanveer Ahmed",
  "phone": "9876543210",
  "email": "tanveer@ptexports.com",
  "role": "managing_director",
  "is_active": true
}
```

POST /auth/change-password

Request

```
{ "current_password": ".....", "new_password": "....." }
```

Response 204 — no body.

Two exceptions

The following routes are **deliberately unauthenticated** — a supplier's mail client, Amazon SES and Twilio call them, and they carry an opaque per-send token rather than a session:

```
GET /procurement/t/o/{token}.gif
GET /procurement/t/c/{token}
POST /procurement/webhooks/ses
POST /procurement/webhooks/twilio/whatsapp
POST /procurement/webhooks/twilio/voice
```

Do not call these from the frontend. They are documented in §17 for completeness.

3. Conventions

Aspect	Convention
IDs	UUID v4, always as a string in JSON
Timestamps	ISO-8601 with timezone: <code>"2026-09-09T14:32:07.123456+00:00"</code> . Nullable fields are <code>null</code> , never <code>""</code>
Dates	<code>"2026-09-09"</code>
Money	JSON <code>number</code> , two decimal places. Computed server-side — never send a computed total
Quantities	JSON <code>number</code> , three decimal places
Confidence	<code>0.0 – 1.0</code>
Enums	Lowercase snake_case strings
Uploads	<code>multipart/form-data</code> , field name <code>file</code> . Cap <code>MAX_UPLOAD_MB</code> (default 25)
Pagination	<code>limit</code> (default 100) + <code>offset</code> . Responses carry <code>count</code> = rows in this page , not the grand total
Empty collections	<code>[]</code> , never <code>null</code>
Async operations	<code>202 Accepted</code> + a <code>task_id</code> ; poll the matching GET
Streaming	<code>text/event-stream</code> . Frames are <code>data: {json}\n\n</code> ; lines starting with <code>:</code> are keep-alives

Three rules worth stating explicitly:

1. **Derived values are always server-computed.** `total_cost`, `bulk_qty`, `bulk_total`, `garment_fob_price`, `available_qty`, `shortfall_qty`, `subtotal`, `cgst` / `sgst` / `igst`, `total` — send the inputs, render what comes back. Do not compute these client-side; you will drift.
2. **`count` is the page size.** No endpoint in Phase 2 returns a grand total. Paginate by fetching until a short page comes back.
3. **Optimistic locking on every edit.** The BOM and PO editors both require `base_revision`. Always send the revision you rendered.

4. The error contract

4.1 Two shapes

Shape A — Stage-1 upload rejections. A rich, flat diagnostic envelope. `error` is at the top level.

```
{
  "error": "document_validation_failed",
  "reason_code": "not_an_order_sheet",
  "submission_id": "a3f2b8c1-4d5e-6f70-8192-a3b4c5d6e7f8",
  "document_fingerprint": {
    "filename": "packing-list.xlsx",
    "mime": "application/vnd.openxmlformats-officedocument.spreadsheetml.sheet",
    "sha256": "9f86d081884c7d659a2feaa0c55ad015a3bf4f1b2b0b822cd15d6c15b0f00a08",
    "size_bytes": 84213
  },
  "validation": {
    "status": "rejected",
    "reason_code": "not_an_order_sheet",
    "expected_kind": "order_sheet",
    "confidence": 0.31,
    "signals_expected": ["ORDER CONFIRMATION", "TAGLIA", "QTY", "COLORE"],
    "signals_found": ["PACKING LIST", "CARTON", "NET WEIGHT"],
    "closest_client_profile": "BOGGI",
    "method": "llm",
    "suggested_fix": "This looks like a packing list. Upload the order confirmation instead."
  }
}
```

Shape B — everything else. FastAPI's `detail` wrapper, containing either a structured object or a plain string.

```
{ "detail": { "error": "stale_revision", "current_revision": 7 } }
```

```
{ "detail": "BOM not found." }
```

4.2 One parser for both

```
export function parseError(status, body) {
  if (body?.error) {
    return { code: body.error, reason: body.reason_code, detail: body, status };
  }
  const d = body?.detail;
  if (typeof d === "string") return { code: "message", message: d, status };
  if (d && typeof d === "object") return { code: d.error ?? "unknown", detail: d, status };
  return { code: "unknown_error", status };
}
```

4.3 Status codes

Code	Meaning	Frontend action
200	OK	
201	Created	
202	Accepted — queued	Start polling
204	No content	
302	Redirect	Tracking links only
400	Bad request	Invalid tracking signature
401	Unauthenticated	Redirect to login
403	Wrong role	Show who <i>can</i> act — the body often names them
404	Not found	
409	State conflict	See §4.4 — the most important family
413	File too large	Show the cap
415	Unsupported file type	Show accepted types
422	Validation failed	Field-level or document-level
500	Server error	Show <code>request_id</code> for support
503	Dependency unavailable	Virus scanner down — offer retry

Every `500` carries a correlation id:

```
{ "detail": "Internal server error", "request_id": "3f9a2b71-..." }
```

Show that id to the user. It is how support traces the failure in the logs.

4.4 The complete 409 catalogue

error	Endpoint	Meaning	Recovery
duplicate_content	Uploads	These exact bytes already exist elsewhere	Upload the correct distinct file
submission_locked	Uploads	Stage 2 already consumed it	Start a new submission
stale_revision	BOM/PO edit	Someone else edited first	Reload, re-present, retry
bom_locked	BOM edit	Approved/locked BOMs reject edits	Read-only
bom_rejected	BOM edit	Must reopen first	Offer Reopen
cutting_confirmation_required	BOM approve	Not cutting-confirmed	Disable Approve upfront
not_ready_for_review	BOM reject	Wrong state	Drive buttons off status
not_rejected	BOM reopen	Not rejected	Hide Reopen
not_approved	BOM export / PO send	Wrong state	Hide the action
spec_suggestion_unconfirmed	Generate BOM	Spec still suggested	Disable upfront
bom_not_approved	Inventory check	BOM not approved	Disable
no_inventory_check	Generate POs	Check has not run	Offer "run the check"
needs_supplier	PO submit	No supplier assigned	Open the supplier picker
empty_po	PO submit	No lines	Disable Submit
not_submittable	PO submit	Not draft/rejected	Drive off status
not_pending_approval	PO approve/reject	Wrong state	Drive off status
po_locked	PO edit / cancel	Sent POs are frozen	Cancel and re-issue
duplicate_name	Create supplier	Name already exists	Offer the existing supplier

Almost every 409 here is avoidable by driving button state off the status the API already returned.

4.5 The 422 catalogue

error	Where
unknown_bom_item / unknown_po_item	The referenced line is not on this record
unsupported_field	Field not in the editable set
invalid_value	Not coercible to a number
negative_value	Negative quantity or price
reason_required	Reject with an empty reason
name_required	Create supplier with an empty name
unknown_status	Board transition to an unknown rung
(Pydantic default)	Malformed body — detail is an array of field errors

5. The role matrix

MD Managing Director · DM Direct Manager · CM Cutting Manager · HR · VW Viewer

Endpoint group	MD	DM	CM	HR	VW
Submissions + uploads	✓	✓			
Breakdown, attachments, generate BOM	✓	✓	✓	✓	✓
Read/edit a BOM, confirm cutting	✓	✓	✓		
Approve / reject / export a BOM	✓				
Reopen a BOM	✓	✓			
Notifications	✓	✓	✓	✓	✓
Upload a pattern	✓	✓	✓	✓	✓
Admin configuration	✓	✓			
Inventory sync + run check	✓	✓			
Inventory reads	✓	✓			✓
Supplier reads	✓	✓			✓
Supplier create / edit / import	✓	✓			
Supplier deactivate / reactivate	✓				
PO reads	✓	✓			✓
PO generate / edit / submit / send / cancel / acknowledge	✓	✓			
PO approve / reject — leather	✓		✓		
PO approve / reject — accessory & service	✓	✓		✓	
Production board — read	✓	✓			✓
Production board — transition	✓	✓	✓		
Chat	✓	✓	✓	✓	✓

Notes:

- **MD is a superuser** for PO approval — they may approve any PO regardless of type.
- Breakdown, attachment and generate-BOM endpoints are gated only by `get_current_user` — any authenticated non-legacy user reaches them. **The UI should still present them as DM/MD tools.**
- The legacy `employee` role is blocked at the router level for every route in this document.

6. Endpoint index

Stage 1 — Intake (7)

Method	Path	Roles
POST	/procurement/submissions	DM MD
POST	/procurement/submissions/{submission_id}/order-sheet	DM MD
POST	/procurement/upload/order-sheet	DM MD
POST	/procurement/submissions/{submission_id}/spec-sheet	DM MD
POST	/procurement/upload/spec-sheet	DM MD
GET	/procurement/submissions/{submission_id}	DM MD
GET	/procurement/submissions/{submission_id}/documents/{document_id}	DM MD

Stage 2/3 — BOM (23)

Method	Path	Roles
POST	/procurement/submissions/{submission_id}/order-breakdown	any
GET	/procurement/submissions/{submission_id}/order-breakdown	any
POST	/procurement/order-styles/{order_style_id}/attachments	any
POST	/procurement/order-styles/{order_style_id}/generate-bom	any
GET	/procurement/boms/{bom_id}	CM MD DM
PATCH	/procurement/boms/{bom_id}/items	CM MD DM
POST	/procurement/boms/{bom_id}/confirm-cutting	CM MD DM
POST	/procurement/boms/{bom_id}/approve	MD
POST	/procurement/boms/{bom_id}/reject	MD
POST	/procurement/boms/{bom_id}/reopen	DM MD
POST	/procurement/boms/{bom_id}/export	MD
GET	/procurement/notifications	any
GET	/procurement/notifications/stream	any
POST	/procurement/notifications/{notification_id}/open	any
POST	/procurement/patterns	any
PUT	/procurement/admin/dxf-yields/{species}	DM MD
POST	/procurement/admin/fabric-roles	DM MD
GET	/procurement/admin/cost-catalog	DM MD
PUT	/procurement/admin/cost-catalog/{garment_code}	DM MD
GET	/procurement/admin/checks	DM MD
PUT	/procurement/admin/checks/{client_code}	DM MD
GET	/procurement/admin/pom-dictionary	DM MD
POST	/procurement/admin/pom-dictionary	DM MD

Stage 4 — Inventory (7)

Method	Path	Roles
POST	/procurement/inventory/preview	DM MD
POST	/procurement/inventory/commit	DM MD
GET	/procurement/inventory/items	DM MD VW
POST	/procurement/boms/{bom_id}/inventory-check	DM MD
GET	/procurement/boms/{bom_id}/inventory-check	DM MD VW
GET	/procurement/inventory-checks/{check_id}	DM MD VW
GET	/procurement/inventory-checks	DM MD VW

Stage 5 — Supplier PO (25)

Method	Path	Roles
POST	/procurement/suppliers/import/preview	DM MD
POST	/procurement/suppliers/import/commit	DM MD
GET	/procurement/suppliers	DM MD VW
GET	/procurement/suppliers/{supplier_id}	DM MD VW
POST	/procurement/suppliers	DM MD
PATCH	/procurement/suppliers/{supplier_id}	DM MD
DELETE	/procurement/suppliers/{supplier_id}	MD
POST	/procurement/suppliers/{supplier_id}/reactivate	MD
POST	/procurement/boms/{bom_id}/generate-pos	DM MD
GET	/procurement/pos	DM MD VW
GET	/procurement/pos/{po_id}	DM MD VW
PATCH	/procurement/pos/{po_id}/items	DM MD
POST	/procurement/pos/{po_id}/submit	DM MD
POST	/procurement/pos/{po_id}/approve	CM MD DM HR
POST	/procurement/pos/{po_id}/reject	CM MD DM HR
POST	/procurement/pos/{po_id}/send	DM MD
POST	/procurement/pos/{po_id}/cancel	DM MD
POST	/procurement/pos/{po_id}/acknowledge	DM MD
GET	/procurement/t/o/{token}.gif	none
GET	/procurement/t/c/{token}	none
POST	/procurement/webhooks/ses	none
POST	/procurement/webhooks/twilio/whatsapp	none
POST	/procurement/webhooks/twilio/voice	none
GET	/procurement/production-tracking	DM MD VW
POST	/procurement/production-tracking/{tracking_id}/transition	MD DM CM

Intelligence (2)

Method	Path	Roles
POST	/chat	any
POST	/chat/stream	any

Total: 64 endpoints (+ 3 auth, + 2 health).

STAGE 1 — INTAKE

Router: `app/modules/procurement/router.py` · Service: `ProcurementService`

7. Submissions and uploads

7.1 `POST /procurement/submissions`

Open an empty upload batch.

Creates the container that pairs an order sheet with a spec sheet. The client may be unknown at this point — it is derived from the order sheet during Stage 2 — so `client_id` is optional.

Roles: DM · MD

Request body (*optional — may be omitted entirely*)

```
{ "client_id": "7c9e6679-7425-40de-944b-e07fc1f90ae7" }
```

Response `201`

```
{
  "submission_id": "a3f2b8c1-4d5e-6f70-8192-a3b4c5d6e7f8",
  "status": "open"
}
```

Frontend note. You do not have to call this first. The two `/upload/*` endpoints create a submission automatically when a valid file arrives. Use this endpoint when you want to show an empty two-slot workspace before the user picks any file.

7.2 `POST /procurement/submissions/{submission_id}/order-sheet`

7.3 `POST /procurement/submissions/{submission_id}/spec-sheet`

Upload and validate a document into a slot.

Runs the full pipeline: size gate → virus scan → MIME sniff → classification → persist. Accepted documents fill the slot and supersede whatever was there before. Rejected documents still create a `Document` row, so re-uploading the same bytes replays the verdict for free.

Roles: DM · MD

Path: `submission_id` (UUID)

Query: `force` (boolean, default `false`) — accept a `needs_manual_review` document anyway. **Hard rejections are never overridable.**

Request: `multipart/form-data`, field `file`

```
POST /api/v1/procurement/submissions/a3f2b8c1-.../order-sheet
Content-Type: multipart/form-data; boundary=----X

-----X
Content-Disposition: form-data; name="file"; filename="ORDER BOGGI SS27.xlsx"
Content-Type: application/vnd.openxmlformats-officedocument.spreadsheetml.sheet

<bytes>
-----X--
```

Response 201

```
{
  "submission_id": "a3f2b8c1-4d5e-6f70-8192-a3b4c5d6e7f8",
  "document": {
    "id": "b1c2d3e4-5f60-7182-93a4-b5c6d7e8f901",
    "kind": "order_sheet",
    "filename": "ORDER BOGGI SS27.xlsx",
    "mime": "application/vnd.openxmlformats-officedocument.spreadsheetml.sheet",
    "sha256": "e3b0c44298fc1c149afb4c8996fb92427ae41e4649b934ca495991b7852b855",
    "size_bytes": 92416,
    "page_count": null,
    "storage_url": "s3://kairox-docs/submissions/a3f2b8c1/e3b0c442.xlsx",
    "validation": {
      "status": "accepted",
      "classified_as": "order_sheet",
      "spec_type": null,
      "client_match": "BOGGI",
      "confidence": 0.94,
      "method": "heuristic",
      "llm_label": null,
      "llm_model": null,
      "signals_matched": ["ORDER CONFIRMATION", "TAGLIA", "COLORE", "QTY"],
      "signals_expected": ["ORDER CONFIRMATION", "TAGLIA", "QTY", "COLORE"],
      "signals_found": ["ORDER CONFIRMATION", "TAGLIA", "COLORE", "QTY", "STAGIONE"],
      "reason_code": null,
      "suggested_fix": null
    },
    "scan_status": "clean"
  },
  "submission": {
    "order_sheet": { "present": true, "validation_status": "accepted" },
    "spec_sheet": { "present": false, "validation_status": null },
    "complete": false,
    "ready_for_stage_2": false,
    "blocking": ["spec_sheet missing"]
  }
}
```

Errors

Code	reason_code	When
413	file_too_large	Over <code>MAX_UPLOAD_MB</code>
415	unsupported_mime	Not an accepted type
422	empty_or_corrupt	Unparseable
422	virus_detected	Infected — also writes an audit row
422	not_an_order_sheet / not_a_spec_sheet	Wrong document kind
422	wrong_slot	Right kind, wrong slot
422	needs_manual_review	Inconclusive — retry with <code>?force=true</code>
409	duplicate_content	These bytes already exist elsewhere
409	submission_locked	Stage 2 already consumed this submission
503	scanner_unavailable	Virus scanner down — retry
404	—	Submission not found

A **422 rejection** (shape A):

```
{
  "error": "document_validation_failed",
  "reason_code": "not_an_order_sheet",
  "submission_id": "a3f2b8c1-4d5e-6f70-8192-a3b4c5d6e7f8",
  "document_fingerprint": {
    "filename": "packing-list.xlsx",
    "mime": "application/vnd.openxmlformats-officedocument.spreadsheetml.sheet",
    "sha256": "9f86d081884c7d659a2feaa0c55ad015a3bf4f1b2b0b822cd15d6c15b0f00a08",
    "size_bytes": 84213
  },
  "validation": {
    "status": "rejected",
    "reason_code": "not_an_order_sheet",
    "expected_kind": "order_sheet",
    "confidence": 0.31,
    "signals_expected": ["ORDER CONFIRMATION", "TAGLIA", "QTY", "COLORE"],
    "signals_found": ["PACKING LIST", "CARTON", "NET WEIGHT"],
    "closest_client_profile": "BOGGI",
    "method": "llm",
    "suggested_fix": "This looks like a packing list. Upload the order confirmation instead."
  }
}
```

A **409 duplicate_content**:

```
{
  "error": "duplicate_content",
  "submission_id": "a3f2b8c1-4d5e-6f70-8192-a3b4c5d6e7f8",
  "document_fingerprint": {
    "filename": "ORDER BOGGI SS27.xlsx",
    "mime": "application/vnd.openxmlformats-officedocument.spreadsheetml.sheet",
    "sha256": "e3b0c44298fc1c149afb4c8996fb92427ae41e4649b934ca495991b7852b855",
    "size_bytes": 92416
  },
  "conflict": {
    "attempted_slot": "spec_sheet",
    "existing_kind": "order_sheet",
    "existing_validation_status": "accepted",
    "same_submission": true,
    "existing_submission_id": "a3f2b8c1-4d5e-6f70-8192-a3b4c5d6e7f8",
    "suggested_fix": "This file is already accepted as the other slot of this submission; upload the correct,"
  }
}
```

Frontend notes

- The `validation` block is the whole rejection screen. Render `signals_expected` and `signals_found` **side by side** — the difference is the explanation.
- Show a **Force accept** button *only* when `reason_code == "needs_manual_review"`.
- Re-uploading an identical file is safe and cheap — it replays the cached verdict.
- `blocking` is a plain-language checklist. Render it directly.

7.4 POST /procurement/upload/order-sheet

7.5 POST /procurement/upload/spec-sheet

Upload without an existing submission.

Identical to 7.2/7.3 except that a **new submission is created only if the file passes validation**. A rejected file returns diagnostics with no submission created.

Roles: DM · MD · **Query:** `force` · **Request:** `multipart/form-data`, field `file`

Response `201` — identical shape to 7.2, with a freshly minted `submission_id`.

Important: capture the returned `submission_id` and use it for the paired upload. Otherwise you create two orphaned single-slot submissions.

7.6 GET /procurement/submissions/{submission_id}

Submission status and the Stage-2 readiness gate.

Poll this after each upload to drive the workspace.

Roles: DM · MD

Response 200

```
{
  "order_sheet": { "present": true, "validation_status": "accepted" },
  "spec_sheet": { "present": true, "validation_status": "accepted" },
  "complete": true,
  "ready_for_stage_2": true,
  "blocking": []
}
```

Not ready:

```
{
  "order_sheet": { "present": true, "validation_status": "accepted" },
  "spec_sheet": { "present": false, "validation_status": "needs_manual_review" },
  "complete": false,
  "ready_for_stage_2": false,
  "blocking": ["spec_sheet needs_manual_review"]
}
```

`ready_for_stage_2` is true when **both** slots are `accepted` and neither has a blocking `scan_status`. `blocking` also carries entries like `"order_sheet scan_status=infected"`.

Errors: 404 submission not found.

7.7 GET /procurement/submissions/{submission_id}/documents/{document_id}

The full validation report for one document.

Use it to show details for a superseded document, or to re-open a rejection without re-uploading.

Roles: DM · MD

Response 200

```
{
  "submission_id": "a3f2b8c1-4d5e-6f70-8192-a3b4c5d6e7f8",
  "document": {
    "id": "b1c2d3e4-5f60-7182-93a4-b5c6d7e8f901",
    "kind": "order_sheet",
    "filename": "ORDER BOGGI SS27.xlsx",
    "mime": "application/vnd.openxmlformats-officedocument.spreadsheetml.sheet",
    "sha256": "e3b0c44298fc1c149afb4c8996fb92427ae41e4649b934ca495991b7852b855",
    "size_bytes": 92416,
    "page_count": null,
    "storage_url": "s3://kairox-docs/submissions/a3f2b8c1/e3b0c442.xlsx",
    "validation": {
      "status": "accepted",
      "classified_as": "order_sheet",
      "spec_type": null,
      "client_match": "BOGGI",
      "confidence": 0.94,
      "method": "heuristic",
      "llm_label": null,
      "llm_model": null,
      "signals_matched": ["ORDER CONFIRMATION", "TAGLIA", "COLORE", "QTY"],
      "signals_expected": ["ORDER CONFIRMATION", "TAGLIA", "QTY", "COLORE"],
      "signals_found": ["ORDER CONFIRMATION", "TAGLIA", "COLORE", "QTY", "STAGIONE"],
      "reason_code": null,
      "suggested_fix": null
    },
    "scan_status": "clean"
  }
}
```

Errors: 404 — document not found, or it does not belong to this submission.

STAGE 2/3 — BOM

Router: `app/modules/bom/router.py` · Services: `BomService`, `NotificationService`

8. Breakdown and style attachment

8.1 `POST /procurement/submissions/{submission_id}/order-breakdown`

Queue extraction of the order document into a style list.

Returns immediately. AI extraction takes 30–200 seconds and runs in a Celery worker. Atomically claims the submission so two clicks cannot both enqueue.

Roles: any authenticated user

Request: no body

Response `202` — freshly queued

```
{
  "submission_id": "a3f2b8c1-4d5e-6f70-8192-a3b4c5d6e7f8",
  "status": "queued",
  "task_id": "c7f21a90-3b4c-5d6e-7f80-91a2b3c4d5e6"
}
```

Response `202` — already extracted

```
{ "submission_id": "a3f2b8c1-...", "status": "already_ready", "task_id": null }
```

Response `202` — a worker is mid-flight

```
{ "submission_id": "a3f2b8c1-...", "status": "already_processing", "task_id": null }
```

Errors

Code	Body	When
<code>409</code>	<code>"submission not ready: order document not accepted, or already claimed"</code>	The order slot is not <code>accepted</code> , or someone else claimed it
<code>404</code>	—	Submission not found

Frontend handling

<code>status</code>	Do
<code>queued</code>	Start polling 8.2 every 3–5s
<code>already_processing</code>	Start polling 8.2 — someone else started it
<code>already_ready</code>	Fetch 8.2 once; do not poll

8.2 GET /procurement/submissions/{submission_id}/order-breakdown

Read the breakdown, or its progress.

Roles: any authenticated user

Response 200 — ready

```
{
  "submission_id": "a3f2b8c1-4d5e-6f70-8192-a3b4c5d6e7f8",
  "status": "ready",
  "styles": [
    {
      "id": "d4e5f607-1829-3a4b-5c6d-7e8f90a1b2c3",
      "style_signature": "CLERMONT",
      "style_name": "CLERMONT",
      "material": "SHEEP GLASS",
      "qty": 60,
      "per_size_qty": { "46": 10, "48": 20, "50": 20, "52": 10 },
      "colors": [
        {
          "color_key": "BLACK",
          "color_label": "NERO / BLACK",
          "qty": 40,
          "per_size_qty": { "46": 6, "48": 14, "50": 14, "52": 6 },
          "warnings": []
        },
        {
          "color_key": "COGNAC",
          "color_label": "COGNAC",
          "qty": 20,
          "per_size_qty": { "46": 4, "48": 6, "50": 6, "52": 4 },
          "warnings": []
        }
      ],
      "warnings": [],
      "spec_document_id": "e5f60718-293a-4b5c-6d7e-8f90a1b2c3d4",
      "spec_match_status": "suggested",
      "pattern_reference_id": "f6071829-3a4b-5c6d-7e8f-90a1b2c3d4e5",
      "dxf_match_status": "suggested",
      "bom_id": null
    },
    {
      "id": "07182930-4b5c-6d7e-8f90-a1b2c3d4e5f6",
      "style_signature": "CARNABY",
      "style_name": "CARNABY",
      "material": "GOAT SUEDE",
      "qty": 40,
      "per_size_qty": { "48": 15, "50": 15, "52": 10 },
      "colors": [
        {
          "color_key": "TAUPE",
          "color_label": "TAUPE",
          "qty": 40,
          "per_size_qty": { "48": 15, "50": 15, "52": 10 },
          "warnings": []
        }
      ]
    }
  ]
}
```

```

    ],
    "warnings": ["size_column_ambiguous"],
    "spec_document_id": null,
    "spec_match_status": "none",
    "pattern_reference_id": "1829304b-5c6d-7e8f-90a1-b2c3d4e5f607",
    "dxf_match_status": "confirmed",
    "bom_id": null
  }
],
"warnings": []
}

```

Response 200 — still working

```
{ "submission_id": "a3f2b8c1-...", "status": "processing", "styles": [], "warnings": [] }
```

Response 200 — nothing yet

```
{ "submission_id": "a3f2b8c1-...", "status": "not_started", "styles": [], "warnings": [] }
```

Field reference

Field	Meaning
<code>style_signature</code>	The cross-order-stable key. Drives the DCM memory
<code>qty</code> / <code>per_size_qty</code>	Style totals. <code>per_size_qty</code> sums to <code>qty</code>
<code>colors[]</code>	The colour dimension. Each colour's <code>per_size_qty</code> sums to its own <code>qty</code> ; the colours sum to the style
<code>spec_match_status</code>	<code>none</code> <code>suggested</code> <code>confirmed</code> — a <code>suggested</code> spec blocks BOM generation
<code>dxf_match_status</code>	Same vocabulary. Never blocks — a missing DXF only lowers DCM confidence
<code>warnings[]</code>	Extraction uncertainties, e.g. <code>size_column_ambiguous</code> , <code>qty_mismatch</code>
<code>bom_id</code>	Null until generated — this is your "done" signal when polling generation

8.3 POST /procurement/order-styles/{order_style_id}/attachments

Confirm or override the suggested spec and pattern for one style.

Synchronous — two column writes, no AI. This is the human click that turns `suggested` into `confirmed`.

Roles: any authenticated user

Request — accept both current suggestions

```
{}
```

Request — override the spec with a specific document

```
{ "spec_document_id": "aabbccdd-1122-3344-5566-778899aabbcc" }
```

Request — detach the spec

```
{ "clear_spec": true }
```

Request — a mixed action

```
{
  "spec_document_id": "aabbccdd-1122-3344-5566-778899aabbcc",
  "clear_dxf": true
}
```

Body reference

Field	Type	Effect
<code>spec_document_id</code>	UUID?	Bind this document → <code>confirmed</code>
<code>pattern_reference_id</code>	UUID?	Bind this pattern → <code>confirmed</code>
<code>clear_spec</code>	bool	Detach → <code>none</code>
<code>clear_dxf</code>	bool	Detach → <code>none</code>

Rules

- An explicit id **overrides** any suggestion and confirms it.
- Omitting an id **accepts the current suggestion** (if any) and confirms it.
- `clear_x` and `x_id` together is a contradiction → `422`.

Response `200` — the full updated style, identical in shape to one element of 8.2's `styles[]`:

```
{
  "id": "d4e5f607-1829-3a4b-5c6d-7e8f90a1b2c3",
  "style_signature": "CLERMONT",
  "style_name": "CLERMONT",
  "material": "SHEEP GLASS",
  "qty": 60,
  "per_size_qty": { "46": 10, "48": 20, "50": 20, "52": 10 },
  "colors": [ /* ... */ ],
  "warnings": [],
  "spec_document_id": "aabbccdd-1122-3344-5566-778899aabbcc",
  "spec_match_status": "confirmed",
  "pattern_reference_id": null,
  "dxf_match_status": "none",
  "bom_id": null
}
```

Errors

Code	error	When
<code>404</code>	<code>order_style_not_found</code>	Unknown style
<code>422</code>	(Pydantic)	<code>clear_spec</code> with <code>spec_document_id</code> , or <code>clear_dxf</code> with <code>pattern_reference_id</code>

8.4 POST /procurement/order-styles/{order_style_id}/generate-bom

Queue BOM generation for one confirmed style.

Preconditions are checked **before** queuing, so you get a 4xx immediately rather than a task failure later.

Roles: any authenticated user · **Request:** no body

Response 202 — queued

```
{
  "order_style_id": "d4e5f607-1829-3a4b-5c6d-7e8f90a1b2c3",
  "status": "queued",
  "bom_id": null,
  "task_id": "9a8b7c6d-5e4f-3021-a1b2-c3d4e5f60718"
}
```

Response 202 — already generated (idempotent replay)

```
{
  "order_style_id": "d4e5f607-...",
  "status": "already_generated",
  "bom_id": "11223344-5566-7788-99aa-bbccddeeff00",
  "task_id": null
}
```

Errors

Code	error	When	Fix
404	order style not found	Unknown style	—
422	spec suggestion is unconfirmed – confirm or clear it before generating	spec_match_status == "suggested"	Call 8.3 first

The rule: a style may be generated with **no** spec (none → the BOM carries a spec_pending warning). It may **not** be generated with a spec nobody confirmed.

Polling for completion. There is no dedicated status endpoint. Poll 8.2 and watch for bom_id on that style to become non-null.

9. The BOM tree and its lifecycle

9.1 GET /procurement/boms/{bom_id}

The editable BOM tree.

Roles: Cutting Manager · MD · DM

Response 200

```
{
  "id": "11223344-5566-7788-99aa-bbccddeeff00",
  "status": "draft",
  "revision": 1,
  "currency": "USD",
  "order_qty": 60,
  "garment_fob_price": 107.25,
  "bulk_total": 6435.00,
  "cutting_confirmed_at": null,
  "approved_at": null,
  "rejection_reason": null,
  "export_document_id": null,
  "exported_at": null,
  "items": [
    {
      "id": "aa000001-0000-0000-0000-000000000001",
      "category": "main_material",
      "name": "SHEEP GLASS",
      "material_color": "BLACK",
      "qty_per_garment": 34.500,
      "uom": "dm2",
      "unit_price": 1.80,
      "bulk_qty": 2070.000,
      "total_cost": 62.10,
      "dcm_source": "template",
      "dcm_confidence": 0.9500,
      "annotation": null
    },
    {
      "id": "aa000001-0000-0000-0000-000000000002",
      "category": "sub_material",
      "name": "GOAT SUEDE",
      "material_color": "BLACK",
      "qty_per_garment": 2.600,
      "uom": "dm2",
      "unit_price": 2.10,
      "bulk_qty": 156.000,
      "total_cost": 5.46,
      "dcm_source": "ai_estimate",
      "dcm_confidence": 0.5000,
      "annotation": "estimated from POM area - confirm at cutting"
    },
    {
      "id": "aa000001-0000-0000-0000-000000000003",
      "category": "lining",
      "name": "VISCOSE LINING",
      "material_color": "BLACK",
      "qty_per_garment": 1.200,
      "uom": "mtr",

```

```
"unit_price": 3.40,
"bulk_qty": 72.000,
"total_cost": 4.08,
"dcm_source": "similar_style",
"dcm_confidence": 0.7000,
"annotation": null
},
{
  "id": "aa000001-0000-0000-0000-000000000004",
  "category": "thread",
  "name": "POLY THREAD 40/2",
  "material_color": "BLACK",
  "qty_per_garment": 120.000,
  "uom": "mtr",
  "unit_price": 0.01,
  "bulk_qty": 7200.000,
  "total_cost": 1.20,
  "dcm_source": null,
  "dcm_confidence": null,
  "annotation": null
},
{
  "id": "aa000001-0000-0000-0000-000000000005",
  "category": "accessory",
  "name": "YKK ZIP #5 60CM",
  "material_color": "BLACK",
  "qty_per_garment": 1.000,
  "uom": "pcs",
  "unit_price": 1.41,
  "bulk_qty": 60.000,
  "total_cost": 1.41,
  "dcm_source": null,
  "dcm_confidence": null,
  "annotation": null
},
{
  "id": "aa000001-0000-0000-0000-000000000006",
  "category": "manufacturing",
  "name": "CUTTING + STITCHING",
  "material_color": null,
  "qty_per_garment": 1.000,
  "uom": null,
  "unit_price": 30.00,
  "bulk_qty": 60.000,
  "total_cost": 30.00,
  "dcm_source": null,
  "dcm_confidence": null,
  "annotation": null
},
{
```

```

    "id": "aa000001-0000-0000-0000-000000000007",
    "category": "packaging",
    "name": "POLYBAG + CARTON",
    "material_color": null,
    "qty_per_garment": 1.000,
    "uom": null,
    "unit_price": 3.00,
    "bulk_qty": 60.000,
    "total_cost": 3.00,
    "dcm_source": null,
    "dcm_confidence": null,
    "annotation": null
  },
  {
    "id": "aa000001-0000-0000-0000-000000000008",
    "category": "fob_charge",
    "name": "FOB CHARGE",
    "material_color": null,
    "qty_per_garment": 1.000,
    "uom": null,
    "unit_price": 5.00,
    "bulk_qty": 60.000,
    "total_cost": 5.00,
    "dcm_source": null,
    "dcm_confidence": null,
    "annotation": null
  }
]
}

```

Field reference

Field	Meaning
revision	Send this as base_revision on every edit
order_qty	Units ordered — the multiplier for every bulk figure
garment_fob_price	Σ of every line's total_cost — the per-garment FOB price
bulk_total	Σ of every line's bulk amount = order_qty × garment_fob_price
qty_per_garment	For material lines this IS the DCM
bulk_qty	order_qty × qty_per_garment — this is what the inventory check calls required_qty
total_cost	qty_per_garment × unit_price — per-garment, not bulk
dcm_source	template dxf similar_style ai_estimate provisional manual . Null for non-material lines
dcm_confidence	0–1. Highlight anything below 0.70
annotation	Free text from the generator — usually why an estimate was used

Errors: 404 — "BOM not found."

9.2 PATCH /procurement/boms/{bom_id}/items

Bulk-edit BOM lines under an optimistic revision lock.

Roles: Cutting Manager · MD · DM

Request

```
{
  "base_revision": 1,
  "edits": [
    { "bom_item_id": "aa000001-0000-0000-0000-000000000002", "field": "dcm",          "value": 3.1 },
    { "bom_item_id": "aa000001-0000-0000-0000-000000000003", "field": "unit_price", "value": 3.6 }
  ]
}
```

Body reference

Field	Type	Notes
base_revision	int ≥ 1	The revision you rendered
edits	array, min 1	
edits[].bom_item_id	UUID	Must belong to this BOM
edits[].field	string	Only dcm , qty_per_garment , unit_price
edits[].value	number	Must be ≥ 0

dcm and qty_per_garment are the same column — dcm is the friendlier name for material lines.

Response 200

```

{
  "revision": 2,
  "reconfirm_required": false,
  "recomputed": {
    "id": "11223344-5566-7788-99aa-bbccddeeff00",
    "status": "draft",
    "revision": 2,
    "currency": "USD",
    "order_qty": 60,
    "garment_fob_price": 108.25,
    "bulk_total": 6495.00,
    "cutting_confirmed_at": null,
    "approved_at": null,
    "rejection_reason": null,
    "export_document_id": null,
    "exported_at": null,
    "items": [
      {
        "id": "aa000001-0000-0000-0000-000000000002",
        "category": "sub_material",
        "name": "GOAT SUEDE",
        "material_color": "BLACK",
        "qty_per_garment": 3.100,
        "uom": "dm2",
        "unit_price": 2.10,
        "bulk_qty": 186.000,
        "total_cost": 6.51,
        "dcm_source": "manual",
        "dcm_confidence": 1.0000,
        "annotation": "estimated from POM area - confirm at cutting"
      }
    ]
  }
}

```

Three behaviours to build for:

1. **Editing a quantity on a material line stamps** `dcm_source: "manual"` , **confidence** `1.0000` . The badge changes colour in the response — render the returned tree, do not patch locally.
2. `reconfirm_required: true` means the BOM had been cutting-confirmed and your quantity edit **revoked it**. The status in `recomputed` drops to `"draft"` . Show a prominent banner.
3. `recomputed` **is the complete new tree**. Replace your local state with it.

Errors

Code	error	Extra	Meaning
409	stale_revision	current_revision	Reload and retry
409	bom_locked		Approved/locked — read-only
409	bom_rejected		Reopen it first
422	unknown_bom_item	bom_item_id	Not on this BOM
422	unsupported_field	field	Not in the editable set
422	invalid_value	bom_item_id, field, value	Not a number
422	negative_value	bom_item_id, field, value	Below zero
404			BOM not found

```
{ "detail": { "error": "stale_revision", "current_revision": 3 } }
```

9.3 POST /procurement/boms/{bom_id}/confirm-cutting

The cutting-manager gate — and the learning step.

Moves the BOM to `ready_for_review`, **writes every confirmed material quantity into the consumption memory**, records DXF yield observations, and notifies the MD and DM.

Roles: Cutting Manager (MD/DM bypass as superusers) · **Request:** no body

Response 200

```
{
  "bom_id": "11223344-5566-7788-99aa-bbccddeeff00",
  "status": "ready_for_review",
  "cutting_confirmed_at": "2026-09-09T11:42:03.881204+00:00",
  "templates_backfilled": 3,
  "notifications_created": 2
}
```

Field	Meaning
templates_backfilled	Material lines written into the DCM memory. Surface this — it is the system getting smarter
notifications_created	Usually 2 (MD + DM). Each carries its own 2-hour escalation clock

Errors

Code	error	Extra
409	invalid_state_for_confirmation	current_status — only <code>draft</code> or <code>ready_for_review</code> are allowed
404		BOM not found

9.4 POST /procurement/boms/{bom_id}/approve

MD approval — the moment the order becomes real.

Materialises the Client → Order → Style → SKU hierarchy, advances the production board, and **runs the inventory check**.

Roles: MD only

Request *(optional)*

```
{ "lock": false }
```

`lock: true` sets status to `locked` instead of `approved`. Functionally identical; `locked` states intent more strongly.

Response `200`

```
{
  "bom_id": "11223344-5566-7788-99aa-bbccddeeff00",
  "status": "approved",
  "inventory_check_id": "cc001122-3344-5566-7788-99aabbccddee"
}
```

`inventory_check_id` may be `null` if the check failed. The BOM is still approved — the check is best-effort. Offer a "run the check" action when it is null.

Errors

Code	error	Meaning
409	cutting_confirmation_required	cutting_confirmed_at is null. Disable Approve upfront
404		BOM not found

```
{
  "detail": {
    "error": "cutting_confirmation_required",
    "message": "The cutting manager must confirm the BOM before approval."
  }
}
```

Frontend note. `inventory_check_id` is your navigation target — route straight to the check result. This is the natural hand-off point in the UI.

9.5 `POST /procurement/boms/{bom_id}/reject`

MD rejection with a mandatory reason.

Roles: MD only

Request

```
{ "reason": "Goat suede DCM looks 40% high against the CR1-02F5 template. Please re-check." }
```

Response `200`

```
{
  "bom_id": "11223344-5566-7788-99aa-bbccddeeff00",
  "status": "rejected",
  "rejection_reason": "Goat suede DCM looks 40% high against the CR1-02F5 template. Please re-check."
}
```

Errors

Code	error	Extra
422	reason_required	Empty or whitespace-only
409	bom_locked	An approved BOM cannot be rejected
409	not_ready_for_review	current_status
404		BOM not found

9.6 POST /procurement/boms/{bom_id}/reopen

Reopen a rejected BOM for editing.

Bumps the revision and **clears the cutting confirmation** — the whole chain restarts properly.

Roles: DM · MD · **Request:** no body

Response 200

```
{
  "bom_id": "11223344-5566-7788-99aa-bbccddeeff00",
  "status": "draft",
  "revision": 2
}
```

Errors: 409 not_rejected (with current_status) · 404

Frontend note. Warn the user that reopening clears the cutting confirmation and the BOM will need re-confirming.

9.7 POST /procurement/boms/{bom_id}/export

Render and store the BOM as a PDF.

Deduplicated by sha256. Re-exporting an already-exported BOM returns the existing document rather than making a second one.

Roles: MD only · **Request:** no body

Response 200

```
{
  "bom_id": "11223344-5566-7788-99aa-bbccddeeff00",
  "status": "exported",
  "export_document_id": "dd112233-4455-6677-8899-aabbccddeeff",
  "sha256": "5891b5b522d5df086d0ff0b110fbd9d21bb4fc7163af34d08286a2e846f6be03",
  "mime": "application/pdf",
  "storage_url": "s3://kairox-docs/exports/11223344/5891b5b5.pdf"
}
```

A note on `mime` . The renderer falls back WeasyPrint → ReportLab → raw HTML, so `mime` can be `text/html` and the extension `.html` if neither PDF library is available. **Branch on `mime` ; do not assume PDF.**

Errors: `409 not_approved` · `404`

10. Notifications

10.1 `GET /procurement/notifications`

The caller's notification rows.

Roles: any authenticated user · **Query:** `unread_only` (bool, default `false`)

Response `200`

```
{
  "notifications": [
    {
      "id": "ee223344-5566-7788-99aa-bbccddeeff00",
      "type": "bom_review",
      "channel": "in_app",
      "subject": "BOM ready for review - CLERMONT (BOG-SS27-001)",
      "body": "The cutting manager has confirmed the BOM for CLERMONT. Please review and approve.",
      "entity_type": "bom",
      "entity_id": "11223344-5566-7788-99aa-bbccddeeff00",
      "status": "sent",
      "scheduled_for": "2026-09-09T13:42:03.881204+00:00",
      "sent_at": "2026-09-09T11:42:04.102331+00:00",
      "opened_at": null
    },
    {
      "id": "ff334455-6677-8899-aabb-ccddee0011",
      "type": "po_awaiting_approval",
      "channel": "in_app",
      "subject": "PO draft awaiting cross-check",
      "body": "A supplier PO is awaiting your cross-check approval. http://localhost:3000/pos/...",
      "entity_type": "purchase_order",
      "entity_id": "99887766-5544-3322-1100-ffeeddccbbaa",
      "status": "pending",
      "scheduled_for": "2026-09-09T15:10:00.000000+00:00",
      "sent_at": null,
      "opened_at": null
    }
  ],
  "unread": 2
}
```

Field	Meaning
type	bom_review · po_awaiting_approval · po_dispatch · po_escalation_exhausted
entity_type + entity_id	Your deep link
scheduled_for	When the email escalation fires if nobody opens it
opened_at	Non-null Cancels the escalation
unread	Badge count

10.2 GET /procurement/notifications/stream

Server-Sent Events stream of the caller's notifications.

Emits the unseen backlog on connect (marking each `sent`), then polls every 10 seconds, sending `: keep-alive` comment frames when idle.

Roles: any authenticated user · **Response:** `200 text/event-stream`

```
data: {"id":"ee223344-...", "type":"bom_review", "channel":"in_app", "subject":"BOM ready for review - CLERMONT", "
: keep-alive

: keep-alive

data: {"id":"ff334455-...", "type":"po_awaiting_approval", ...}
```

```
const es = new EventSource("/api/v1/procurement/notifications/stream",
                          { withCredentials: true });
es.onmessage = (e) => addNotification(JSON.parse(e.data));
es.onerror    = () => { es.close(); startPolling(); };
```

Frames are the same objects 10.1 returns. **The table is the source of truth** — a dropped connection loses nothing; poll 10.1 to resync.

10.3 POST /procurement/notifications/{notification_id}/open

Mark a notification opened — this cancels the email escalation.

Roles: any authenticated user (recipient only) · **Request:** no body

Response 200

```
{
  "id": "ee223344-5566-7788-99aa-bbccddeeff00",
  "type": "bom_review",
  "channel": "in_app",
  "subject": "BOM ready for review - CLERMONT (BOG-SS27-001)",
  "body": "The cutting manager has confirmed the BOM for CLERMONT. Please review and approve.",
  "entity_type": "bom",
  "entity_id": "11223344-5566-7788-99aa-bbccddeeff00",
  "status": "opened",
  "scheduled_for": "2026-09-09T13:42:03.881204+00:00",
  "sent_at": "2026-09-09T11:42:04.102331+00:00",
  "opened_at": "2026-09-09T12:01:55.400129+00:00"
}
```

Errors: 403 — not the recipient · 404

Do not skip this call. It is the only thing that stops the escalation email. Fire it when the user actually views the notification.

11. Patterns

11.1 POST /procurement/patterns

Upload a DXF cutting pattern for background parsing.

Stored immediately; parsing (pieces, areas, fabrics, sizes) runs in Celery.

Roles: any authenticated user

Query: `style_signature` (string, optional) — link the pattern to a style

Request: `multipart/form-data`, field `file` (a `.dxf`)

Response 202

```
{
  "job_id": "7a6b5c4d-3e2f-1009-8877-665544332211",
  "channel": "pattern:CLERMONT"
}
```

`channel` is a suggested subscription key if you wire realtime push. There is no polling endpoint for the job itself — the parsed pattern surfaces as a `pattern_reference_id` suggestion on the next breakdown, and improves DCM confidence on the next BOM generation.

12. Admin configuration

Six configuration sets, hot-editable. Every write also refreshes the in-memory `config_store` snapshot, so changes take effect **without a restart**.

Roles for all of §12: DM · MD

12.1 PUT /procurement/admin/dxf-yields/{species}

Set the leather yield factor for a species.

`species` ∈ `sheep` · `goat` · `calf` · `_default`

Request

```
{ "factor": 1.18, "note": "Revised after Q2 2026 cutting-floor observations" }
```

Response 200

```
{ "species": "sheep", "factor": 1.18 }
```

12.2 POST /procurement/admin/fabric-roles

Map a raw DXF fabric label to a BOM role.

Request

```
{
  "label": "PELLE PRINCIPALE",
  "role": "main",
  "category": "main_material",
  "is_leather": true
}
```

Response **200** — echoes the body:

```
{
  "label": "PELLE PRINCIPALE",
  "role": "main",
  "category": "main_material",
  "is_leather": true
}
```

12.3 GET /procurement/admin/cost-catalog

The default cost lines per garment code.

Response **200**

```
{
  "JACKET": [
    { "category": "manufacturing", "name": "CUTTING + STITCHING", "uom": null, "unit_price": 30.00, "qty_pe
    { "category": "packaging", "name": "POLYBAG + CARTON", "uom": null, "unit_price": 3.00, "qty_pe
    { "category": "fob_charge", "name": "FOB CHARGE", "uom": null, "unit_price": 5.00, "qty_pe
  ],
  "VEST": [
    { "category": "manufacturing", "name": "CUTTING + STITCHING", "uom": null, "unit_price": 18.00, "qty_pe
    { "category": "packaging", "name": "POLYBAG", "uom": null, "unit_price": 1.50, "qty_pe
  ]
}
```

12.4 PUT /procurement/admin/cost-catalog/{garment_code}

Replace the whole cost catalogue for a garment code. `garment_code` is uppercased server-side.

Request

```
{
  "lines": [
    { "category": "manufacturing", "name": "CUTTING + STITCHING", "uom": null, "unit_price": 32.00, "qty_per_
    { "category": "packaging",      "name": "POLYBAG + CARTON",      "uom": null, "unit_price": 3.20, "qty_per_
    { "category": "fob_charge",      "name": "FOB CHARGE",            "uom": null, "unit_price": 5.00, "qty_per_
  ]
}
```

Field	Type	Notes
<code>category</code>	string	A <code>BomItemCategory</code> value
<code>name</code>	string	The BOM line label
<code>uom</code>	string?	
<code>unit_price</code>	number?	≥ 0
<code>qty_per_garment</code>	number	> 0 , default <code>1</code>

Response **200**

```
{ "garment_code": "JACKET", "lines": 3 }
```

This is a full replace, not a merge. Send every line you want to keep.

12.5 GET /procurement/admin/checks

Per-client BOM sanity rules.

Response **200**

```
[
  {
    "client_code": "BOGGI",
    "rules": [
      { "id": "dcm-range-main", "kind": "range", "severity": "warn", "field": "main_material.qty_per_garment
      { "id": "fob-floor",      "kind": "range", "severity": "error", "field": "garment_fob_price",
    ]
  }
]
```

12.6 PUT /procurement/admin/checks/{client_code}

Replace the whole rule set for a client.

Request

```
{
  "rules": [
    { "id": "dcm-range-main", "kind": "range", "severity": "warn", "field": "main_material.qty_per_garment",
    { "id": "fob-floor", "kind": "range", "severity": "error", "field": "garment_fob_price",
  ]
}
```

Field	Type	Notes
<code>id</code>	string	Stable rule id
<code>kind</code>	string	Rule family, e.g. <code>range</code>
<code>severity</code>	string	<code>warn</code> (default) <code>error</code>
<code>field</code>	string?	Dotted path to the checked value
<code>range</code>	<code>[lo, hi]</code> ?	For range rules
<code>params</code>	object?	Rule-specific extras

Response 200

```
{ "client_code": "BOGGI", "rules": 2 }
```

12.7 GET /procurement/admin/pom-dictionary

Every native-term → standard-POM mapping.

Response 200

```
[
  { "source_term": "torace", "pom_code": "CHEST", "language": "it", "garment_type_id": null, "weight": 1
  { "source_term": "胸囲", "pom_code": "CHEST", "language": "ja", "garment_type_id": null, "weight": 1
  { "source_term": "spalle", "pom_code": "SHOULDER", "language": "it", "garment_type_id": null, "weight": 1
  { "source_term": "袖ぐり", "pom_code": "SHOULDER", "language": "ja", "garment_type_id": "aa11bb22-cc33-dd44"
]
```

12.8 POST /procurement/admin/pom-dictionary

Add or update one mapping.

Request

```
{
  "source_term": "lunghezza",
  "pom_code": "BODY_LENGTH",
  "language": "it",
  "garment_type_code": "JACKET",
  "weight": 1
}
```

Field	Type	Notes
source_term	string	The native term as it appears on spec sheets
pom_code	string	The standard code
language	string?	Inferred from the term if omitted (CJK → ja)
garment_type_code	string?	Null applies to any garment type
weight	int	Default 1 . Higher wins on ambiguity

Response 200

```
{
  "source_term": "lunghezza",
  "pom_code": "BODY_LENGTH",
  "language": "it",
  "garment_type_id": "aa11bb22-cc33-dd44-ee55-ff6677889900"
}
```

garment_type_id is null if garment_type_code was omitted or did not resolve.

STAGE 4 — INVENTORY

Router: `app/modules/inventory/router.py` · Service: `InventoryService`

13. Stock master

13.1 `POST /procurement/inventory/preview`

Dry-run a stock spreadsheet. Writes nothing.

Always call this before 13.2 and show the result.

Roles: DM · MD · **Request:** `multipart/form-data` , field `file` (`.xlsx`)

Response `200`

```

{
  "raw_count": 412,
  "kept": 389,
  "dropped": 23,
  "warnings": [
    "row 118: blank description - dropped",
    "row 204: qty could not be parsed ('N/A') - treated as 0",
    "12 rows merged into 5 keys by normalization"
  ],
  "rows": [
    {
      "normalized_key": "sheep glass black",
      "description": "SHEEP GLASS BLACK",
      "uom": "dm2",
      "qty_on_hand": 3400.0,
      "rate": 1.75,
      "color": "BLACK",
      "lots": 3
    },
    {
      "normalized_key": "goat suede black",
      "description": "GOAT SUEDE BLACK",
      "uom": "dm2",
      "qty_on_hand": 100.0,
      "rate": 2.05,
      "color": "BLACK",
      "lots": 1
    },
    {
      "normalized_key": "ykk zip 5 60cm black",
      "description": "YKK ZIP #5 60CM BLACK",
      "uom": "pcs",
      "qty_on_hand": 240.0,
      "rate": 1.35,
      "color": "BLACK",
      "lots": 2
    }
  ]
}

```

Field	Meaning
<code>raw_count</code>	Rows read from the sheet
<code>kept</code> / <code>dropped</code>	After normalisation and validation
<code>warnings[]</code>	Human-readable, row-numbered. Render all of them
<code>rows[].normalized_key</code>	The matching key. Multiple sheet rows can collapse into one
<code>rows[].lots</code>	How many sheet rows merged into this key

13.2 POST /procurement/inventory/commit

Apply the spreadsheet to the stock master.

Idempotent upsert on `normalized_key`. **The sheet wins on quantity.** Keys absent from the sheet are **soft-deactivated**, never deleted.

Roles: DM · MD · **Request:** `multipart/form-data`, field `file`

Response `200` — the preview block plus two counters:

```
{
  "raw_count": 412,
  "kept": 389,
  "dropped": 23,
  "warnings": ["row 118: blank description - dropped"],
  "rows": [ /* ... same shape as preview ... */ ],
  "committed": 389,
  "deactivated": 7
}
```

Field	Meaning
<code>committed</code>	Rows inserted or updated
<code>deactivated</code>	Existing keys absent from this sheet , now <code>is_active: false</code>

Warn about `deactivated` before the user commits. A partial sheet will silently deactivate everything it omits. This is the single most destructive action in Stage 4.

13.3 GET /procurement/inventory/items

The paged stock master.

Roles: DM · MD · Viewer

Query: `search` (string, matches description/key) · `limit` (default 100) · `offset` (default 0)

Response `200`

```
{
  "items": [
    {
      "id": "b0001111-2222-3333-4444-555566667777",
      "description": "SHEEP GLASS BLACK",
      "normalized_key": "sheep glass black",
      "uom": "dm2",
      "qty_on_hand": 3400.0,
      "rate": 1.75,
      "color": "BLACK",
      "is_active": true
    },
    {
      "id": "b0002222-3333-4444-5555-666677778888",
      "description": "GOAT SUEDE BLACK",
      "normalized_key": "goat suede black",
      "uom": "dm2",
      "qty_on_hand": 100.0,
      "rate": 2.05,
      "color": "BLACK",
      "is_active": true
    }
  ],
  "count": 2
}
```

count is the rows in this page, not the grand total. Paginate until a short page arrives.

Note: this endpoint returns **qty_on_hand** only. Reserved and available are computed **per BOM** inside a check — there is no global "available" figure.

14. Inventory checks

14.1 POST /procurement/boms/{bom_id}/inventory-check

Run (or re-run) the stock check for an approved BOM.

Releases this BOM's prior reservations, re-matches every stockable line, and re-reserves. One transaction, candidate rows locked.

Roles: DM · MD · **Request:** no body

Response 200

```
{
  "inventory_check_id": "cc001122-3344-5566-7788-99aabbccdde",
  "bom_id": "11223344-5566-7788-99aa-bbccddeeff00",
  "status": "complete",
  "run_at": "2026-09-09T12:15:44.220118+00:00",
  "summary": {
    "badge": "out_of_stock",
    "lines_total": 6,
    "sufficient": 3,
    "partial": 1,
    "out_of_stock": 2,
    "flags": { "unmatched": 1, "uom_mismatch": 0 },
    "shortfall_value": 1247.60,
    "currency": "INR"
  },
  "lines": [
    {
      "bom_item_id": "aa000001-0000-0000-0000-000000000001",
      "category": "main_material",
      "name": "SHEEP GLASS",
      "material_color": "BLACK",
      "required_qty": 2070.0,
      "uom": "dm2",
      "matched": {
        "inventory_item_id": "b0001111-2222-3333-4444-555566667777",
        "description": "SHEEP GLASS BLACK",
        "method": "key",
        "uom": "dm2"
      },
      "on_hand_qty": 3400.0,
      "available_qty": 2400.0,
      "reserved_for_this_bom": 2070.0,
      "shortfall_qty": 0.0,
      "status": "sufficient",
      "flags": [],
      "suggestion": null
    },
    {
      "bom_item_id": "aa000001-0000-0000-0000-000000000002",
      "category": "sub_material",
      "name": "GOAT SUEDE",
      "material_color": "BLACK",
      "required_qty": 156.0,
      "uom": "dm2",
      "matched": {
        "inventory_item_id": "b0002222-3333-4444-5555-666677778888",
        "description": "GOAT SUEDE BLACK",
        "method": "key",
        "uom": "dm2"
      }
    }
  ]
}
```

```

    },
    "on_hand_qty": 100.0,
    "available_qty": 100.0,
    "reserved_for_this_bom": 100.0,
    "shortfall_qty": 56.0,
    "status": "partial",
    "flags": [],
    "suggestion": null
  },
  {
    "bom_item_id": "aa000001-0000-0000-0000-000000000003",
    "category": "lining",
    "name": "VISCOSE LINING",
    "material_color": "BLACK",
    "required_qty": 72.0,
    "uom": "mtr",
    "matched": null,
    "on_hand_qty": 0.0,
    "available_qty": 0.0,
    "reserved_for_this_bom": 0.0,
    "shortfall_qty": 72.0,
    "status": "out_of_stock",
    "flags": ["suggestion", "unmatched"],
    "suggestion": {
      "inventory_item_id": "b0003333-4444-5555-6666-777788889999",
      "description": "VISCOSE LINING FABRIC BLACK",
      "score": 0.67
    }
  },
  {
    "bom_item_id": "aa000001-0000-0000-0000-000000000004",
    "category": "thread",
    "name": "POLY THREAD 40/2",
    "material_color": "BLACK",
    "required_qty": 7200.0,
    "uom": "mtr",
    "matched": {
      "inventory_item_id": "b0004444-5555-6666-7777-888899990000",
      "description": "POLYESTER THREAD 40/2 BLACK",
      "method": "alias",
      "uom": "mtr"
    },
    "on_hand_qty": 50000.0,
    "available_qty": 44000.0,
    "reserved_for_this_bom": 7200.0,
    "shortfall_qty": 0.0,
    "status": "sufficient",
    "flags": [],
    "suggestion": null
  },

```

```
{
  "bom_item_id": "aa000001-0000-0000-0000-000000000005",
  "category": "accessory",
  "name": "YKK ZIP #5 60CM",
  "material_color": "BLACK",
  "required_qty": 60.0,
  "uom": "pcs",
  "matched": {
    "inventory_item_id": "b0005555-6666-7777-8888-99990000aaaa",
    "description": "YKK ZIP #5 60CM BLACK",
    "method": "key",
    "uom": "pcs"
  },
  "on_hand_qty": 240.0,
  "available_qty": 240.0,
  "reserved_for_this_bom": 60.0,
  "shortfall_qty": 0.0,
  "status": "sufficient",
  "flags": [],
  "suggestion": null
},
{
  "bom_item_id": "aa000001-0000-0000-0000-000000000007",
  "category": "packaging",
  "name": "POLYBAG + CARTON",
  "material_color": null,
  "required_qty": 60.0,
  "uom": null,
  "matched": null,
  "on_hand_qty": 0.0,
  "available_qty": 0.0,
  "reserved_for_this_bom": 0.0,
  "shortfall_qty": 60.0,
  "status": "out_of_stock",
  "flags": ["unmatched"],
  "suggestion": null
}
],
"excluded": [
  {
    "bom_item_id": "aa000001-0000-0000-0000-000000000006",
    "name": "CUTTING + STITCHING",
    "category": "manufacturing"
  },
  {
    "bom_item_id": "aa000001-0000-0000-0000-000000000008",
    "name": "FOB CHARGE",
    "category": "fob_charge"
  }
]
```

```
]
}
```

Summary reference

Field	Meaning
<code>badge</code>	The worst line status. <code>out_of_stock</code> > <code>partial</code> > <code>sufficient</code>
<code>lines_total</code>	Stockable lines checked (excludes <code>excluded[]</code>)
<code>flags</code>	How many lines carry each flag
<code>shortfall_value</code>	Σ (shortfall $\times$ stock rate). Always INR — independent of the BOM's currency

Line reference

Field	Meaning
<code>required_qty</code>	The BOM line's <code>bulk_qty</code> — <code>order_qty $\times$ qty_per_garment</code>
<code>matched</code>	The primary matched stock row, or <code>null</code> . <code>method</code> $\in$ <code>key</code> <code>alias</code> <code>manual</code>
<code>on_hand_qty</code>	Summed across all matched lots, unit-converted
<code>available_qty</code>	<code>on_hand</code> – reservations held by other BOMs
<code>reserved_for_this_bom</code>	What this check just claimed = <code>min(required, available)</code>
<code>shortfall_qty</code>	<code>required – reserved</code> . This is exactly what Stage 5 orders
<code>flags[]</code>	<code>unmatched</code> · <code>uom_mismatch</code> · <code>suggestion</code>
<code>suggestion</code>	A fuzzy candidate — advisory only, never applied

`excluded[]` lists `manufacturing` and `fob_charge` lines. **Always render it**, greyed with an explanation, or the user will wonder why lines are missing.

Errors

Code	error	Meaning
409	<code>bom_not_approved</code>	Only <code>approved</code> / <code>locked</code> / <code>exported</code> BOMs can be checked
404		BOM not found

14.2 GET /procurement/inventory-checks/{check_id}

14.3 GET /procurement/boms/{bom_id}/inventory-check

Read a stored check. 14.2 by check id; 14.3 fetches the latest for a BOM.

Roles: DM · MD · Viewer

Response `200` — the same envelope as 14.1, with one difference:

`available_qty` equals `on_hand_qty` on a stored read. The live available figure is not persisted; `reserved_for_this_bom` is recovered as `required – shortfall`. Only a fresh run (14.1) returns the true live `available_qty`.

Show the `run_at` timestamp so the user knows how fresh the data is, and offer a "re-run" action.

Errors: 404 — no check found for this id/BOM.

14.4 GET /procurement/inventory-checks

The grouped dashboard: client → order → style.

Roles: DM · MD · Viewer · **Query:** client_id (UUID?) · order_id (UUID?)

Response 200

```
{
  "clients": [
    {
      "client_id": "7c9e6679-7425-40de-944b-e07fc1f90ae7",
      "client_name": "BOGGI MILANO",
      "orders": [
        {
          "client_order_id": "3a4b5c6d-7e8f-9001-1223-3445566778899",
          "order_number": "BOG-SS27-001",
          "styles": [
            {
              "style_id": "4b5c6d7e-8f90-0112-2334-4556677889900",
              "style_name": "CLERMONT",
              "bom_id": "11223344-5566-7788-99aa-bbccddeeff00",
              "inventory_check_id": "cc001122-3344-5566-7788-99aabbccddee",
              "badge": "out_of_stock",
              "shortfall_lines": 3,
              "checked_at": "2026-09-09T12:15:44.220118+00:00"
            },
            {
              "style_id": "5c6d7e8f-9001-1223-3445-566778899001",
              "style_name": "CARNABY",
              "bom_id": "22334455-6677-8899-aabb-ccddee0011",
              "inventory_check_id": "dd112233-4455-6677-8899-aabbccddee",
              "badge": "sufficient",
              "shortfall_lines": 0,
              "checked_at": "2026-09-09T12:22:10.554902+00:00"
            }
          ]
        }
      ]
    }
  ],
  "totals": {
    "boms_checked": 2,
    "fully_sufficient": 1,
    "with_shortfall": 1
  }
}
```

A client whose id could not be resolved appears with `client_id: null` and `client_name: null` under the key `"unknown"`.

STAGE 5 — SUPPLIER PO

Router: `app/modules/supplier_po/router.py` · Services: `PoService` , `SupplierService` , `ProductionTrackingService`

15. Suppliers

15.1 `POST /procurement/suppliers/import/preview`

Dry-run a supplier spreadsheet.

Parses both the contact sheet and the provision (purchase) ledger. Writes nothing.

Roles: DM · MD · **Request:** `multipart/form-data` , field `file`

Response `200`

```
{
  "contact_rows": 84,
  "provision_rows": 1362,
  "suppliers": 79,
  "suppliers_with_phone": 71,
  "suppliers_with_email": 44,
  "history_rows": 612,
  "warnings": [
    "row 42: supplier name blank - skipped",
    "8 provision rows had no parseable date - first/last purchase left null"
  ],
  "sample_history": [
    {
      "supplier": "S.N. TRADERS",
      "article": "sheep glass black",
      "mode": "leather",
      "txn_count": 14,
      "last_rate": 1.72
    },
    {
      "supplier": "AL-AMEEN LEATHERS",
      "article": "goat suede",
      "mode": "leather",
      "txn_count": 9,
      "last_rate": 2.01
    },
    {
      "supplier": "ZIP WORLD",
      "article": "ykk zip 5 60cm",
      "mode": "accessory",
      "txn_count": 22,
      "last_rate": 1.30
    }
  ]
}
```

`sample_history` holds at most **25** rows — a preview, not the full set.

15.2 POST /procurement/suppliers/import/commit

Apply the supplier spreadsheet.

Three actions: upsert suppliers (**enrich, never blank** — an existing phone or email is not wiped by an omitting sheet), **rebuild the supply history wholesale**, and soft-deactivate suppliers absent from the sheet.

Roles: DM · MD · **Request:** `multipart/form-data`, field `file`

Response `200` — the preview block plus three counters:

```
{
  "contact_rows": 84,
  "provision_rows": 1362,
  "suppliers": 79,
  "suppliers_with_phone": 71,
  "suppliers_with_email": 44,
  "history_rows": 612,
  "warnings": [],
  "sample_history": [ /* ... */ ],
  "committed": 79,
  "deactivated": 3
}
```

The supply history is **replaced entirely**, not merged. A partial sheet destroys matching evidence for everything it omits.

15.3 GET /procurement/suppliers

The supplier directory.

Roles: DM · MD · Viewer

Query: `q` (search) · `service` (filter) · `active` (bool) · `limit` (100) · `offset` (0)

Response `200`

```

{
  "suppliers": [
    {
      "id": "e0001111-2222-3333-4444-555566667777",
      "name": "S.N. TRADERS",
      "phone": "+919840012345",
      "email": "sales@sntraders.in",
      "service": "Leather supply",
      "gstn": "33AABCS1429B1ZP",
      "address": "12 Anna Salai, Chennai 600002",
      "currency": "INR",
      "payment_terms_days": 60,
      "lead_time_days": 10,
      "is_active": true,
      "email_status": "valid",
      "supplier_type": "leather",
      "state_code": "33",
      "whatsapp_phone": "+919840012345",
      "has_contact": true
    },
    {
      "id": "e0002222-3333-4444-5555-666677778888",
      "name": "ZIP WORLD",
      "phone": "+912266778899",
      "email": null,
      "service": "Zips and trims",
      "gstn": "27AACZW1234K1Z5",
      "address": "Andheri East, Mumbai 400069",
      "currency": "INR",
      "payment_terms_days": 45,
      "lead_time_days": 7,
      "is_active": true,
      "email_status": "unknown",
      "supplier_type": "accessory",
      "state_code": "27",
      "whatsapp_phone": "+912266778899",
      "has_contact": true
    }
  ],
  "count": 2
}

```

Field	Meaning
supplier_type	Drives PO approval routing — leather → Cutting Manager; else MD/DM/HR
state_code	First two GSTIN digits. Drives intra vs inter-state GST
email_status	invalid (bounced) means email is skipped entirely on send
has_contact	email OR phone . False → any PO for them is no_contact_channel

15.4 GET /procurement/suppliers/{supplier_id}

One supplier with their purchase history and open POs.

Roles: DM · MD · Viewer

Response 200

```

{
  "id": "e0001111-2222-3333-4444-555566667777",
  "name": "S.N. TRADERS",
  "phone": "+919840012345",
  "email": "sales@sntraders.in",
  "service": "Leather supply",
  "gstn": "33AABCS1429B1ZP",
  "address": "12 Anna Salai, Chennai 600002",
  "currency": "INR",
  "payment_terms_days": 60,
  "lead_time_days": 10,
  "is_active": true,
  "email_status": "valid",
  "supplier_type": "leather",
  "state_code": "33",
  "whatsapp_phone": "+919840012345",
  "has_contact": true,
  "supply_history": [
    {
      "normalized_description": "sheep glass black",
      "raw_description": "SHEEP GLASS BLACK 0.6-0.8MM",
      "mode": "leather",
      "uom": "dm2",
      "txn_count": 14,
      "last_purchased_at": "2026-07-11",
      "last_rate": 1.72,
      "min_rate": 1.58,
      "max_rate": 1.89
    },
    {
      "normalized_description": "goat suede black",
      "raw_description": "GOAT SUEDE BLACK",
      "mode": "leather",
      "uom": "dm2",
      "txn_count": 6,
      "last_purchased_at": "2026-05-02",
      "last_rate": 2.05,
      "min_rate": 1.95,
      "max_rate": 2.20
    }
  ],
  "open_pos": [
    { "id": "99887766-5544-3322-1100-ffeeddccbbaa", "po_number": "PO-07(25-26)", "status": "sent", "total": 5
  ]
}

```

supply_history is the evidence behind supplier ranking — render it on the supplier-selection panel.

Errors: 404 — "Supplier not found."

15.5 POST /procurement/suppliers

Create a supplier. The "none of these — add a new vendor" path from the PO supplier picker.

Roles: DM · MD

Request

```
{
  "name": "NEW LEATHER CO",
  "phone": "+919000011111",
  "email": "sales@newleather.in",
  "service": "Leather supply",
  "gstn": "33AAACN1234A1Z9",
  "address": "45 Periamet, Chennai 600003",
  "currency": "INR",
  "payment_terms_days": 45,
  "lead_time_days": 12,
  "supplier_type": "leather",
  "whatsapp_phone": "+919000011111"
}
```

Only `name` is required. Defaults applied server-side: `currency` "INR", `payment_terms_days` 60, `lead_time_days` 10. `state_code` is derived from the GSTIN.

Response 201 — the full supplier block (same shape as 15.3's rows).

Errors: 422 `name_required` · 409 `duplicate_name` (with `name`)

15.6 PATCH /procurement/suppliers/{supplier_id}

Update a supplier. Send only the fields you are changing — omitted fields are untouched.

Roles: DM · MD

Request

```
{ "email": "purchase@sntraders.in", "supplier_type": "leather" }
```

Accepts the same fields as 15.5, all optional.

Response 200 — the updated supplier block.

Errors: 404

15.7 DELETE /procurement/suppliers/{supplier_id}

Soft-deactivate. Never a hard delete — history is preserved.

Roles: MD only

Response 200

```
{ "id": "e0002222-3333-4444-5555-666677778888", "is_active": false }
```

Errors: 404

15.8 POST /procurement/suppliers/{supplier_id}/reactivate

Reverse a deactivation.

Roles: MD only · Request: no body

Response 200

```
{ "id": "e0002222-3333-4444-5555-666677778888", "is_active": true }
```

Errors: 404

16. Purchase orders

16.1 POST /procurement/boms/{bom_id}/generate-pos

Turn the inventory shortfall into supplier POs.

Matches each shortfall line to a supplier, groups by supplier into one PO each, and holds unmatched lines as `needs_supplier` drafts.

Roles: DM · MD · Request: no body

Response 200

```
{
  "bom_id": "11223344-5566-7788-99aa-bbccddeeff00",
  "resolved": 1,
  "needs_supplier": 1,
  "purchase_orders": [
    {
      "id": "99887766-5544-3322-1100-ffeeddccbbaa",
      "po_number": null,
      "status": "draft",
      "revision": 1,
      "supplier_id": "e0001111-2222-3333-4444-555566667777",
      "bom_id": "11223344-5566-7788-99aa-bbccddeeff00",
      "client_order_id": "3a4b5c6d-7e8f-9001-1223-3445566778899",
      "buyer_ref": "#BOG-SS27-001",
      "issue_date": null,
      "delivery_days": 10,
      "payment_terms_days": 60,
      "currency": "INR",
      "gst_mode": "INTRA",
      "subtotal": 114.80,
      "cgst": 6.89,
      "sgst": 6.89,
      "igst": 0.0,
      "round_off": 0.42,
      "total": 129.00,
      "needs_supplier": false,
      "no_contact_channel": false,
      "match_method": "ledger",
      "candidates": {
        "ranked": [
          {
            "supplier_id": "e0001111-2222-3333-4444-555566667777",
            "supplier_name": "S.N. TRADERS",
            "score": 0.91,
            "txn_count": 6,
            "has_contact": true,
            "last_rate": 2.05,
            "last_purchased_at": "2026-05-02"
          },
          {
            "supplier_id": "e0003333-4444-5555-6666-777788889999",
            "supplier_name": "AL-AMEEN LEATHERS",
            "score": 0.64,
            "txn_count": 3,
            "has_contact": true,
            "last_rate": 2.18,
            "last_purchased_at": "2025-11-20"
          }
        ]
      }
    }
  ]
}
```

```
,
  "approved_at": null,
  "rejected_at": null,
  "rejection_reason": null,
  "sent_at": null,
  "pdf_document_id": null,
  "first_opened_at": null,
  "first_clicked_at": null,
  "current_rung": 0,
  "next_escalation_at": null,
  "acknowledged_at": null,
  "acknowledged_channel": null,
  "items": [
    {
      "id": "f0001111-2222-3333-4444-555566667777",
      "item_no": 1,
      "description": "GOAT SUEDE",
      "color": "BLACK",
      "uom": "dm2",
      "qty": 56.0,
      "unit_price": 2.05,
      "amount": 114.80,
      "inventory_item_id": "b0002222-3333-4444-5555-666677778888",
      "bom_item_id": "aa000001-0000-0000-0000-000000000002"
    }
  ],
  "supplier": {
    "id": "e0001111-2222-3333-4444-555566667777",
    "name": "S.N. TRADERS",
    "email": "sales@sntraders.in",
    "phone": "+919840012345",
    "gst_in": "33AABCS1429B1ZP",
    "address": "12 Anna Salai, Chennai 600002",
    "supplier_type": "leather",
    "email_status": "valid"
  }
},
{
  "id": "88776655-4433-2211-00ff-eeddcbbaa99",
  "po_number": null,
  "status": "draft",
  "revision": 1,
  "supplier_id": null,
  "bom_id": "11223344-5566-7788-99aa-bbccddeeff00",
  "client_order_id": "3a4b5c6d-7e8f-9001-1223-3445566778899",
  "buyer_ref": "#BOG-SS27-001",
  "issue_date": null,
  "delivery_days": 10,
  "payment_terms_days": 60,
  "currency": "INR",
```

```
"gst_mode": "INTER",
"subtotal": 244.80,
"cgst": 0.0,
"sgst": 0.0,
"igst": 29.38,
"round_off": -0.18,
"total": 274.00,
"needs_supplier": true,
"no_contact_channel": false,
"match_method": null,
"candidates": {
  "method": "fuzzy",
  "ranked": [
    {
      "supplier_id": "e0004444-5555-6666-7777-888899990000",
      "supplier_name": "TEXTILE HOUSE",
      "score": 0.41,
      "txn_count": 2,
      "has_contact": true,
      "last_rate": 3.30,
      "last_purchased_at": "2025-08-14"
    },
    {
      "supplier_id": "e0005555-6666-7777-8888-99990000aaaa",
      "supplier_name": "CHENNAI LININGS",
      "score": 0.38,
      "txn_count": 5,
      "has_contact": false,
      "last_rate": 3.10,
      "last_purchased_at": "2025-06-02"
    }
  ],
  "suggestion": "TEXTILE HOUSE",
  "ambiguous": true
},
"approved_at": null,
"rejected_at": null,
"rejection_reason": null,
"sent_at": null,
"pdf_document_id": null,
"first_opened_at": null,
"first_clicked_at": null,
"current_rung": 0,
"next_escalation_at": null,
"acknowledged_at": null,
"acknowledged_channel": null,
"items": [
  {
    "id": "f0002222-3333-4444-5555-666677778888",
    "item_no": 1,
```

```

      "description": "VISCOSE LINING",
      "color": "BLACK",
      "uom": "mtr",
      "qty": 72.0,
      "unit_price": 3.40,
      "amount": 244.80,
      "inventory_item_id": null,
      "bom_item_id": "aa000001-0000-0000-0000-000000000003"
    }
  ]
}
]
}

```

Response **200** — already generated (idempotent)

```

{
  "bom_id": "11223344-...",
  "already_generated": true,
  "purchase_orders": [ /* the existing POs */ ]
}

```

Response **200** — nothing to order

```

{
  "bom_id": "11223344-...",
  "purchase_orders": [],
  "message": "No shortfall lines - nothing to order."
}

```

Errors

Code	error	Meaning
409	no_inventory_check	Run the check first
404		BOM not found

Frontend notes

- **needs_supplier: true** is the blocker. Those POs cannot be submitted until a supplier is assigned via 16.4.
- **candidates.ambiguous: true** means the system deliberately refused to choose. Show *"too close to call — please pick one"*.
- **candidates.ranked[]** is your comparison table: score, transaction count, last rate, last purchase date, contactable.
- A PO without a supplier has **no supplier key** in the response — guard for it.
- **match_method** is **ledger** | **alias** | **category** | **null**. Show it as a provenance chip; **category** is a weaker signal than **ledger**.

16.2 GET /procurement/pos

List purchase orders.

Roles: DM · MD · Viewer

Query: `status` · `needs_supplier` (bool) · `bom_id` (UUID) · `supplier_id` (UUID)

Response `200`

```
{
  "purchase_orders": [ /* full PO objects – same shape as 16.1 */ ],
  "count": 2
}
```

Build `?needs_supplier=true` as its own prominent tab. Those are the POs blocking the pipeline.

16.3 GET `/procurement/pos/{po_id}`

One purchase order in full.

Roles: DM · MD · Viewer

Response `200` — the full PO object. A **sent** PO looks like this:

```

{
  "id": "99887766-5544-3322-1100-ffeeddccbbaa",
  "po_number": "PO-07(25-26)",
  "status": "escalated",
  "revision": 3,
  "supplier_id": "e0001111-2222-3333-4444-555566667777",
  "bom_id": "11223344-5566-7788-99aa-bbccddeeff00",
  "client_order_id": "3a4b5c6d-7e8f-9001-1223-3445566778899",
  "buyer_ref": "#BOG-SS27-001",
  "issue_date": "2026-09-09",
  "delivery_days": 10,
  "payment_terms_days": 60,
  "currency": "INR",
  "gst_mode": "INTRA",
  "subtotal": 114.80,
  "cgst": 6.89,
  "sgst": 6.89,
  "igst": 0.0,
  "round_off": 0.42,
  "total": 129.00,
  "needs_supplier": false,
  "no_contact_channel": false,
  "match_method": "ledger",
  "candidates": { "ranked": [ /* ... */ ] },
  "approved_at": "2026-09-09T13:02:11.004000+00:00",
  "rejected_at": null,
  "rejection_reason": null,
  "sent_at": "2026-09-09T13:15:40.220000+00:00",
  "pdf_document_id": "aa998877-6655-4433-2211-00ffeeddccbb",
  "first_opened_at": "2026-09-09T14:02:18.900000+00:00",
  "first_clicked_at": null,
  "current_rung": 1,
  "next_escalation_at": "2026-09-09T23:15:40.220000+00:00",
  "acknowledged_at": null,
  "acknowledged_channel": null,
  "items": [ /* ... */ ],
  "supplier": { /* ... */ }
}

```

Money reference

Field	Meaning
gst_mode	INTRA → render CGST + SGST . INTER → render IGST . Never both
round_off	Can be negative
total	subtotal + taxes + round_off

Tracking reference

Field	Meaning
sent_at	Dispatch time
first_opened_at	The supplier's mail client fetched the pixel
first_clicked_at	They clicked a wrapped link
current_rung	0 email · 1 WhatsApp sent · 2 call placed · 3 exhausted
next_escalation_at	Next rung fires here. Null once acknowledged or exhausted
acknowledged_at / acknowledged_channel	The ladder stopped, and how
no_contact_channel	No email and no phone — a human must handle it

Errors: 404 — "Purchase order not found."

16.4 PATCH /procurement/pos/{po_id}/items

Bulk-edit a PO under an optimistic revision lock.

Same contract as the BOM editor, plus header edits, additions and removals.

Roles: DM · MD

Request — assign a supplier to a needs_supplier PO

```
{
  "base_revision": 1,
  "po_edits": { "supplier_id": "e0004444-5555-6666-7777-888899990000" }
}
```

Request — a mixed edit

```
{
  "base_revision": 2,
  "item_edits": [
    { "po_item_id": "f0001111-2222-3333-4444-555566667777", "field": "unit_price", "value": 1.98 },
    { "po_item_id": "f0001111-2222-3333-4444-555566667777", "field": "qty", "value": 60 }
  ],
  "po_edits": { "delivery_days": 14, "buyer_ref": "#BOG-SS27-001-R2" },
  "add_items": [
    {
      "description": "GOAT SUEDE TRIM",
      "color": "BLACK",
      "uom": "dm2",
      "qty": 12,
      "unit_price": 2.05,
      "bom_item_id": null,
      "inventory_item_id": null
    }
  ],
  "remove_item_ids": ["f0009999-8888-7777-6666-555544443333"]
}
```

Body reference

Field	Type	Notes
base_revision	int ≥ 1	Required
item_edits[]	array	{po_item_id, field, value}
item_edits[].field	string	Only description, color, uom, qty, unit_price
po_edits	object?	Only supplier_id, buyer_ref, delivery_days, payment_terms_days
add_items[]	array	New lines
remove_item_ids[]	UUID array	Lines to drop

All four change groups are optional; send only what you need.

Response 200 — the full recomputed PO object with an incremented revision .

Changing supplier_id re-routes the approver and resets any approval. If the PO was pending_approval or approved , it returns to a state requiring fresh approval. Warn the user.

Errors

Code	error	Extra
409	po_locked	A sent PO is frozen — cancel and re-issue
409	stale_revision	current_revision
422	unknown_po_item	po_item_id
422	unsupported_field	field
422	invalid_value	field
404		PO not found

16.5 POST /procurement/pos/{po_id}/submit

Submit for cross-check approval.

Notifies the routed approvers in-app.

Roles: DM · MD · **Request:** no body

Response 200

```
{
  "po_id": "99887766-5544-3322-1100-ffeeddccbbaa",
  "status": "pending_approval",
  "notifications_created": 2
}
```

Errors

Code	error	Meaning
409	not_submittable	Only <code>draft</code> or <code>rejected</code> (with <code>current_status</code>)
409	needs_supplier	Assign a supplier first
409	empty_po	No lines
404		PO not found

16.6 POST /procurement/pos/{po_id}/approve

Cross-check approval, routed by material type.

Roles: Cutting Manager · MD · DM · HR (*the router admits all four; the service enforces the exact routing*)

Routing: `leather` → Cutting Manager or MD. `accessory` / `service` / unset → MD, DM or HR. **MD may approve anything.**

Request: no body

Response 200

```
{ "po_id": "99887766-5544-3322-1100-ffeeddccbbaa", "status": "approved" }
```

Errors

Code	error	Meaning
409	not_pending_approval	Wrong state (with <code>current_status</code>)
403	wrong_approver	The message names the roles that can
404		PO not found

```
{
  "detail": {
    "error": "wrong_approver",
    "message": "This PO routes to ['cutting_manager', 'managing_director']."
  }
}
```

Frontend note. Parse the role list out of `message` and show "Ask a Cutting Manager or the MD to approve this."

16.7 POST /procurement/pos/{po_id}/reject

Reject with a mandatory reason. Same role routing as 16.6.

Request

```
{ "reason": "Rate is 18% above the last purchase from this vendor. Renegotiate." }
```

Response 200

```
{
  "po_id": "99887766-5544-3322-1100-ffeeddccbbaa",
  "status": "rejected",
  "rejection_reason": "Rate is 18% above the last purchase from this vendor. Renegotiate."
}
```

Errors: 422 reason_required · 409 not_pending_approval · 403 wrong_approver · 404

16.8 POST /procurement/pos/{po_id}/send

Allocate the PO number, render the PDF, dispatch, and start the escalation clock.

Roles: DM · MD · **Request:** no body

Response 200 — emailed

```
{
  "po_id": "99887766-5544-3322-1100-ffeeddccbbaa",
  "po_number": "PO-07(25-26)",
  "status": "sent",
  "channel": "email",
  "no_contact_channel": false,
  "pdf_document_id": "aa998877-6655-4433-2211-00ffeeddccbb"
}
```

Response 200 — no email; the sweeper will WhatsApp them

```
{
  "po_id": "88776655-4433-2211-00ff-eeddccbbaa99",
  "po_number": "PO-08(25-26)",
  "status": "sent",
  "channel": null,
  "no_contact_channel": false,
  "pdf_document_id": "bb887766-5544-3322-1100-ffeeddccbbaa"
}
```

Response 200 — no contact at all

```
{
  "po_id": "77665544-3322-1100-ffee-ddccbbaa9988",
  "po_number": "PO-09(25-26)",
  "status": "sent",
  "channel": null,
  "no_contact_channel": true,
  "pdf_document_id": "cc776655-4433-2211-00ff-eeddccbbaa99"
}
```

Field	Meaning
<code>po_number</code>	Allocated at send , financial-year scoped
<code>channel</code>	<code>"email"</code> if emailed; <code>null</code> otherwise
<code>no_contact_channel</code>	true means nobody was contacted. Show a red banner: this needs a manual call
<code>pdf_document_id</code>	The stored PO PDF

Errors: `409 not_approved` (with `current_status`) · `404`

16.9 POST `/procurement/pos/{po_id}/cancel`

Cancel a pre-send PO.

Roles: DM · MD · **Request:** no body

Response `200`

```
{ "po_id": "99887766-5544-3322-1100-ffeeddccbbaa", "status": "cancelled" }
```

Errors: `409 po_locked` — "A sent PO cannot be cancelled in place." · `404`

16.10 POST `/procurement/pos/{po_id}/acknowledge`

Manually record a supplier's confirmation — and stop the escalation ladder.

Use when the supplier confirms by a route the system cannot see: a phone call the buyer made, an email reply, a WhatsApp to a personal number.

Roles: DM · MD

Request

```
{
  "channel": "call",
  "confirmed_qty": 56,
  "notes": "Confirmed by Rajesh on a call at 16:20. Delivery in 8 days."
}
```

Field	Type	Notes
<code>channel</code>	string	Default <code>"manual"</code> . Typically <code>manual</code> · <code>call</code> · <code>whatsapp</code> · <code>email</code>
<code>confirmed_qty</code>	number?	What the supplier actually committed to
<code>notes</code>	string?	Free text

Response `200` — the full PO object with `status: "confirmed"`, `acknowledged_at` set, `next_escalation_at: null`.

Idempotent — acknowledging an already-acknowledged PO returns it unchanged.

Errors: `404`

17. Tracking and webhooks

These five routes are unauthenticated and are not called by the frontend. A supplier's mail client, Amazon SES and Twilio call them. Documented for completeness and for backend review.

17.1 GET /procurement/t/o/{token}.gif

Email open pixel. Returns a 1×1 GIF (`image/gif`) and stamps `first_opened_at` . Always `200` , even for an unknown token — a tracking pixel must never leak whether a token is real.

17.2 GET /procurement/t/c/{token}

Click redirect. Query: `u` (wrapped target), `s` (HMAC signature). Verifies the signature, records the click, then `302` s to the target. `400` on a bad or missing signature.

17.3 POST /procurement/webhooks/ses

Amazon SES delivery / bounce / complaint events.

```
{
  "eventType": "Bounce",
  "tracking_token": "9f2a7c14e0b34d5f8a1b2c3d4e5f6071",
  "bounce": { "bounceType": "Permanent" }
}
```

Response `200 {"ok": true}` . A hard bounce marks the supplier's email `invalid` , fails the response record, and sets `next_escalation_at` to now — the ladder skips straight to WhatsApp.

17.4 POST /procurement/webhooks/twilio/whatsapp

```
{ "tracking_token": "9f2a7c14e0b34d5f8a1b2c3d4e5f6071", "Body": "Confirmed, dispatching Monday" }
```

Response `200 {"ok": true}` . Acknowledges the PO on channel `whatsapp` .

17.5 POST /procurement/webhooks/twilio/voice

```
{ "tracking_token": "9f2a7c14e0b34d5f8a1b2c3d4e5f6071", "Digits": "1" }
```

Response `200 {"ok": true}` . Acknowledges **only** when `Digits == "1"` .

18. The production board

18.1 GET /procurement/production-tracking

The nine-rung board, one row per (order, style).

Roles: DM · MD · Viewer · **Query:** `client_id` (UUID?) · `order_id` (UUID?)

Response 200

```
{
  "trackers": [
    {
      "id": "d0001111-2222-3333-4444-555566667777",
      "client_order_id": "3a4b5c6d-7e8f-9001-1223-3445566778899",
      "order_number": "BOG-SS27-001",
      "client_name": "BOGGI MILANO",
      "style_id": "4b5c6d7e-8f90-0112-2334-4556677889900",
      "style_name": "CLERMONT",
      "bom_id": "11223344-5566-7788-99aa-bbccddeeff00",
      "status": "po_raised",
      "po_count": 2,
      "po_confirmed_count": 0,
      "material_ready_at": null,
      "released_at": null
    },
    {
      "id": "d0002222-3333-4444-5555-666677778888",
      "client_order_id": "3a4b5c6d-7e8f-9001-1223-3445566778899",
      "order_number": "BOG-SS27-001",
      "client_name": "BOGGI MILANO",
      "style_id": "5c6d7e8f-9001-1223-3445-566778899001",
      "style_name": "CARNABY",
      "bom_id": "22334455-6677-8899-aabb-ccddee0011",
      "status": "material_ready",
      "po_count": 1,
      "po_confirmed_count": 1,
      "material_ready_at": "2026-09-09T16:40:02.118000+00:00",
      "released_at": null
    }
  ],
  "count": 2
}
```

Field	Meaning
status	One of the nine rungs
po_count / po_confirmed_count	A natural progress bar. material_ready is reached when they are equal and non-zero
material_ready_at	Stamped when every PO confirmed
released_at	Stamped on the human go/no-go

18.2 POST /procurement/production-tracking/{tracking_id}/transition

Manually move a tracker. The human go/no-go for released_to_production , and the correction path for a stuck row.

Roles: MD · DM · Cutting Manager

Request

```
{ "status": "released_to_production" }
```

Valid values: `awaiting_bom` · `bom_approved` · `inventory_checked` · `po_raised` · `po_confirmed` · `material_ready` · `released_to_production` · `in_production` · `completed`

Response `200`

```
{ "id": "d0001111-2222-3333-4444-555566667777", "status": "released_to_production" }
```

Errors

Code	Body	Meaning
403	"Only DM/MD/Cutting may transition the board."	Wrong role
422	{"error": "unknown_status", "status": "..."}	Not a valid rung
404	"Tracker not found."	

A manual transition may move backwards. It is the correction path for when the system and reality disagree. Every transition is audited. Confirm backwards moves in the UI.

INTELLIGENCE

Router: `app/modules/intelligence/router.py` · Service: `IntelligenceService`

19. Chat

19.1 `POST /chat`

Ask a question about the factory. Get the exact answer.

Roles: any authenticated user

Request

```
{ "question": "Is CLERMONT on schedule?", "use_llm": false }
```

Field	Type	Notes
<code>question</code>	string	Natural language
<code>use_llm</code>	bool	Default <code>false</code> . <code>true</code> uses the LangGraph agent if <code>CHAT_MODEL</code> is configured; otherwise it silently falls back to the deterministic router

Response `200`

```
{
  "answer": "CLERMONT is behind schedule. 60 ordered, 22 produced, 38 remaining with 9 working days to the de",
  "tool": "schedule_status",
  "data": {
    "style": "CLERMONT",
    "ordered": 60,
    "produced": 22,
    "remaining": 38,
    "deadline": "2026-09-22",
    "working_days_left": 9,
    "recent_daily_rate": 2.1,
    "required_daily_rate": 4.2,
    "on_track": false,
    "shortfall_days": 9
  }
}
```

Bottleneck example

```
{
  "answer": "The bottleneck is PASTING. 41 pieces are queued there against a throughput of 12/day – a 3.4-day",
  "tool": "bottleneck",
  "data": {
    "stage": "PASTING",
    "queued": 41,
    "throughput_per_day": 12,
    "backlog_days": 3.4,
    "runner_up": { "stage": "LINE_STITCHING", "queued": 14 }
  }
}
```

Overview example

```
{
  "answer": "4 active orders across 3 clients. 7 styles in production, 2 at risk. 1,840 pieces ordered, 612 c",
  "tool": "overview",
  "data": {
    "active_orders": 4,
    "clients": 3,
    "styles_in_production": 7,
    "styles_at_risk": 2,
    "total_ordered": 1840,
    "total_produced": 612
  }
}
```

No match

```
{
  "answer": "I could not match that to a style or a known question. Try naming a style, or ask about the bott",
  "tool": null,
  "data": null
}
```

Field	Meaning
answer	Prose, ready to display
tool	Which tool answered: <code>schedule_status</code> · <code>bottleneck</code> · <code>plan</code> · <code>overview</code> · <code>search_workflow_docs</code> · <code>null</code>
data	The exact numbers. Render as a structured card beneath the prose — this is the trustworthy part

19.2 POST /chat/stream

The same answer, streamed word-by-word for a typing effect.

Roles: any authenticated user · Request: identical to 19.1 · Response: 200 text/event-stream

```
data: {"delta":"CLERMONT "}

data: {"delta":"is "}

data: {"delta":"behind "}

data: {"delta":"schedule. "}

data: {"done":true,"tool":"schedule_status","data":{"style":"CLERMONT","ordered":60,"produced":22,"remaining":10}}
```

```
async function askStreaming(question, onDelta, onDone) {
  const res = await fetch("/api/v1/chat/stream", {
    method: "POST",
    headers: { "Content-Type": "application/json", Authorization: `Bearer ${token}` },
    body: JSON.stringify({ question, use_llm: false }),
  });
  const reader = res.body.getReader();
  const decoder = new TextDecoder();
  let buffer = "";
  for (;;) {
    const { done, value } = await reader.read();
    if (done) break;
    buffer += decoder.decode(value, { stream: true });
    const frames = buffer.split("\n\n");
    buffer = frames.pop();
    for (const frame of frames) {
      if (!frame.startsWith("data: ")) continue; // skip keep-alive comments
      const payload = JSON.parse(frame.slice(6));
      if (payload.done) onDone(payload); else onDelta(payload.delta);
    }
  }
}
```

Note: `EventSource` cannot POST, so use `fetch` + a stream reader here. The notification stream (10.2) is a GET and works with `EventSource`.

APPENDICES

29. The mock data pack

Every ID below is used consistently across every example in this document. Drop these into your mock server and **one order flows end to end**.

29.1 The scenario

BOGGI MILANO orders 100 leather garments for SS27 across two styles. **CLERMONT** (60 units) is short on goat suede and lining. **CARNABY** (40 units) is fully in stock. CLERMONT generates two POs — one resolved to S.N. TRADERS, one held for a supplier decision.

29.2 Identity constants

```
export const IDS = {
  // people
  user_md: "9e1c4a70-0b2d-4c8e-9f11-6a2b3c4d5e6f",
  user_dm: "8d0b3960-1a2c-3b4d-5e6f-708192a3b4c5",
  user_cutting: "7c9a2850-0912-2a3b-4c5d-6e7f8091a2b3",

  // client & order
  client: "7c9e6679-7425-40de-944b-e07fc1f90ae7", // BOGGI MILANO
  client_order: "3a4b5c6d-7e8f-9001-1223-3445566778899", // BOG-SS27-001

  // stage 1
  submission: "a3f2b8c1-4d5e-6f70-8192-a3b4c5d6e7f8",
  doc_order_sheet: "b1c2d3e4-5f60-7182-93a4-b5c6d7e8f901",
  doc_spec_sheet: "e5f60718-293a-4b5c-6d7e-8f90a1b2c3d4",

  // stage 2 - styles
  order_style_clermont: "d4e5f607-1829-3a4b-5c6d-7e8f90a1b2c3",
  order_style_carnaby: "07182930-4b5c-6d7e-8f90-a1b2c3d4e5f6",
  pattern_clermont: "f6071829-3a4b-5c6d-7e8f-90a1b2c3d4e5",
  pattern_carnaby: "1829304b-5c6d-7e8f-90a1-b2c3d4e5f607",
  style_clermont: "4b5c6d7e-8f90-0112-2334-4556677889900",
  style_carnaby: "5c6d7e8f-9001-1223-3445-566778899001",

  // stage 2/3 - BOMs
  bom_clermont: "11223344-5566-7788-99aa-bbccddeeff00",
  bom_carnaby: "22334455-6677-8899-aabb-ccddee0011",
  bom_export_doc: "dd112233-4455-6677-8899-aabbccddeeff",

  // BOM items (CLERMONT)
  item_sheep_glass: "aa000001-0000-0000-0000-000000000001",
  item_goat_suede: "aa000001-0000-0000-0000-000000000002",
  item_lining: "aa000001-0000-0000-0000-000000000003",
  item_thread: "aa000001-0000-0000-0000-000000000004",
  item_zip: "aa000001-0000-0000-0000-000000000005",
  item_manufacturing: "aa000001-0000-0000-0000-000000000006",
  item_packaging: "aa000001-0000-0000-0000-000000000007",
  item_fob: "aa000001-0000-0000-0000-000000000008",

  // stage 4
  inv_sheep_glass: "b0001111-2222-3333-4444-555566667777",
  inv_goat_suede: "b0002222-3333-4444-5555-666677778888",
  inv_lining: "b0003333-4444-5555-6666-777788889999",
  inv_thread: "b0004444-5555-6666-7777-888899990000",
  inv_zip: "b0005555-6666-7777-8888-99990000aaaa",
  check_clermont: "cc001122-3344-5566-7788-99aabbccdde",
  check_carnaby: "dd112233-4455-6677-8899-aabbccddeeff",

  // stage 5 - suppliers
```

```

sup_sn_traders: "e0001111-2222-3333-4444-555566667777", // leather, TN (33)
sup_zip_world: "e0002222-3333-4444-5555-666677778888", // accessory, MH (27)
sup_al_ameen: "e0003333-4444-5555-6666-777788889999", // leather
sup_textile_house: "e0004444-5555-6666-7777-888899990000", // accessory
sup_chennai_lining: "e0005555-6666-7777-8888-99990000aaaa", // no contact

// stage 5 - POs
po_resolved: "99887766-5544-3322-1100-ffeeddccbbaa", // S.N. TRADERS
po_needs_supplier: "88776655-4433-2211-00ff-eeddccbbaa99", // held
po_item_suede: "f0001111-2222-3333-4444-555566667777",
po_item_lining: "f0002222-3333-4444-5555-666677778888",
po_pdf_doc: "aa998877-6655-4433-2211-00ffeeddccbb",
tracking_token: "9f2a7c14e0b34d5f8a1b2c3d4e5f6071",

// notifications
notif_bom_review: "ee223344-5566-7788-99aa-bbccddeeff00",
notif_po_approval: "ff334455-6677-8899-aabb-ccddee0011",

// board
track_clermont: "d0001111-2222-3333-4444-555566667777",
track_carnaby: "d0002222-3333-4444-5555-666677778888",
};

```

29.3 The consistent numbers

CLERMONT BOM — order_qty 60, FOB \$107.25, bulk total \$6,435.00

Line	Category	DCM	UOM	Rate	Bulk qty	Line cost	DCM source
SHEEP GLASS	main_material	34.500	dm2	1.80	2070.000	62.10	template (0.95)
GOAT SUEDE	sub_material	2.600	dm2	2.10	156.000	5.46	ai_estimate (0.50)
VISCOSE LINING	lining	1.200	mtr	3.40	72.000	4.08	similar_style (0.70)
POLY THREAD 40/2	thread	120.000	mtr	0.01	7200.000	1.20	—
YKK ZIP #5 60CM	accessory	1.000	pcs	1.41	60.000	1.41	—
CUTTING + STITCHING	manufacturing	1.000	—	30.00	60.000	30.00	—
POLYBAG + CARTON	packaging	1.000	—	3.00	60.000	3.00	—
FOB CHARGE	fob_charge	1.000	—	5.00	60.000	5.00	—

CLERMONT inventory check — badge **out_of_stock**, 6 lines, 3 sufficient · 1 partial · 2 out of stock

Line	Required	On hand	Available	Reserved	Shortfall	Status	Flags
SHEEP GLASS	2070.0	3400.0	2400.0	2070.0	0.0	sufficient	—
GOAT SUEDE	156.0	100.0	100.0	100.0	56.0	partial	—
VISCOSE LINING	72.0	0.0	0.0	0.0	72.0	out_of_stock	suggestion, unmatched
POLY THREAD	7200.0	50000.0	44000.0	7200.0	0.0	sufficient	—
YKK ZIP #5	60.0	240.0	240.0	60.0	0.0	sufficient	—
POLYBAG + CARTON	60.0	0.0	0.0	0.0	60.0	out_of_stock	unmatched

Excluded: CUTTING + STITCHING (manufacturing), FOB CHARGE (fob_charge).

The two POs

PO	Supplier	Lines	GST mode	Subtotal	Tax	Total	Flag
po_resolved	S.N. TRADERS (TN, 33)	GOAT SUEDE 56 × 2.05	INTRA	114.80	CGST 6.89 + SGST 6.89	129.00	—
po_needs_supplier	(none)	VISCOSE LINING 72 × 3.40	INTER	244.80	IGST 29.38	274.00	needs_supplier , ambiguous

Buyer state code is 33 (Tamil Nadu), GST rate 12%.

29.4 A minimal MSW handler set

```
import { http, HttpResponse } from "msw";
import { IDS } from "./ids";
const V1 = "/api/v1";

export const handlers = [
  http.post(`${V1}/auth/login`, () =>
    HttpResponse.json({ access_token: "mock.jwt.token", token_type: "bearer" })),

  http.get(`${V1}/auth/me`, () =>
    HttpResponse.json({ id: IDS.user_md, name: "Tanveer Ahmed",
      phone: "9876543210", email: "tanveer@ptexports.com",
      role: "managing_director", is_active: true })),

  http.post(`${V1}/procurement/submissions`, () =>
    HttpResponse.json({ submission_id: IDS.submission, status: "open" },
      { status: 201 })),

  http.get(`${V1}/procurement/submissions/:id`, () =>
    HttpResponse.json({
      order_sheet: { present: true, validation_status: "accepted" },
      spec_sheet: { present: true, validation_status: "accepted" },
      complete: true, ready_for_stage_2: true, blocking: [] })),

  http.get(`${V1}/procurement/submissions/:id/order-breakdown`, () =>
    HttpResponse.json(BREAKDOWN_READY)), // §8.2

  http.get(`${V1}/procurement/boms/:id`, () =>
    HttpResponse.json(BOM_CLERMONT)), // §9.1

  http.patch(`${V1}/procurement/boms/:id/items`, async ({ request }) => {
    const body = await request.json();
    if (body.base_revision !== 1) {
      return HttpResponse.json(
        { detail: { error: "stale_revision", current_revision: 2 } },
        { status: 409 });
    }
    return HttpResponse.json({ revision: 2, reconfirm_required: false,
      recomputed: BOM_CLERMONT_V2 });
  }),

  http.post(`${V1}/procurement/boms/:id/approve`, () =>
    HttpResponse.json({ bom_id: IDS.bom_clermont, status: "approved",
      inventory_check_id: IDS.check_clermont })),

  http.get(`${V1}/procurement/boms/:id/inventory-check`, () =>
    HttpResponse.json(CHECK_CLERMONT)), // §14.1

  http.post(`${V1}/procurement/boms/:id/generate-pos`, () =>
```

```

    HttpResponse.json(GENERATED_POS)),          // §16.1

    http.get(`${V1}/procurement/pos`, () =>
      HttpResponse.json({ purchase_orders: [PO_RESOLVED, PO_NEEDS_SUPPLIER],
                          count: 2 })),

    http.get(`${V1}/procurement/production-tracking`, () =>
      HttpResponse.json(BOARD)),                // §18.1
  ];

```

29.5 Mocking the async flows

Breakdown — flip status after a few polls so you can build the real spinner:

```

let polls = 0;
http.get(`${V1}/procurement/submissions/:id/order-breakdown`, () => {
  polls += 1;
  if (polls < 4) {
    return HttpResponse.json({ submission_id: IDS.submission,
                              status: "processing", styles: [], warnings: [] });
  }
  return HttpResponse.json(BREAKDOWN_READY);
});

```

Notifications SSE:

```

http.get(`${V1}/procurement/notifications/stream`, () => {
  const stream = new ReadableStream({
    start(controller) {
      const enc = new TextEncoder();
      controller.enqueue(enc.encode(`data: ${JSON.stringify(NOTIF_BOM_REVIEW)}\n\n`));
      const t = setInterval(() => controller.enqueue(enc.encode(": keep-alive\n\n")), 10000);
      setTimeout(() => { clearInterval(t); controller.close(); }, 60000);
    },
  });
  return new HttpResponse(stream, { headers: { "Content-Type": "text/event-stream" } });
});

```

Every error state worth building for:

```

export const ERRORS = {
  staleRevision: [409, { detail: { error: "stale_revision", current_revision: 3 } }],
  bomLocked: [409, { detail: { error: "bom_locked",
    message: "A locked/approved BOM rejects all edits." } }],
  cuttingRequired: [409, { detail: { error: "cutting_confirmation_required",
    message: "The cutting manager must confirm the BOM before approval."
  } }],
  specUnconfirmed: [422, { detail: "spec suggestion is unconfirmed – confirm or clear it before generating
  noInventoryCheck: [409, { detail: { error: "no_inventory_check",
    message: "Run the inventory check before generating supplier POs." } }],
  needsSupplier: [409, { detail: { error: "needs_supplier",
    message: "Assign a supplier before submitting." } }],
  wrongApprover: [403, { detail: { error: "wrong_approver",
    message: "This PO routes to ['cutting_manager', 'managing_director']."
  } }],
  poLocked: [409, { detail: { error: "po_locked",
    message: "A sent PO is frozen – cancel + re-issue." } }],
  docRejected: [422, { error: "document_validation_failed",
    reason_code: "not_an_order_sheet", /* ... §4.1 ... */ }],
  serverError: [500, { detail: "Internal server error",
    request_id: "3f9a2b71-8c4d-4e5f-a012-b3c4d5e6f708" } ]],
};

```

30. Frontend call sequences

30.1 Intake to breakdown

POST	/procurement/submissions	→ submission_id
POST	/procurement/submissions/{id}/order-sheet (multipart)	→ 201 4xx diagnostics
POST	/procurement/submissions/{id}/spec-sheet (multipart)	→ 201 4xx diagnostics
GET	/procurement/submissions/{id}	→ ready_for_stage_2?
	└─ if false: show `blocking`, let the user re-upload	
POST	/procurement/submissions/{id}/order-breakdown	→ 202 queued
GET	/procurement/submissions/{id}/order-breakdown ⌚ 3-5s	→ until status="ready"

30.2 Style to BOM

per style shown in the breakdown:

POST	/procurement/order-styles/{id}/attachments	→ confirm spec + DXF
POST	/procurement/order-styles/{id}/generate-bom	→ 202 queued
GET	/procurement/submissions/{sid}/order-breakdown ⌚ 3-5s	→ until this style's bom_id ≠ null
GET	/procurement/boms/{bom_id}	→ render the editor

30.3 Edit and confirm

```
GET    /procurement/boms/{id}                                → note `revision`
PATCH /procurement/boms/{id}/items {base_revision, edits[]} → new revision + recomputed tree
    └─ 409 stale_revision → GET again, re-present, retry
POST   /procurement/boms/{id}/confirm-cutting                → ready_for_review
```

30.4 Approval to inventory

```
GET    /procurement/notifications                            → the MD sees the review notice
POST   /procurement/notifications/{id}/open                  → cancels the email escalation
GET    /procurement/boms/{id}                                → review
POST   /procurement/boms/{id}/approve {lock:false}          → approved + inventory_check_id
GET    /procurement/inventory-checks/{inventory_check_id}    → the check result
    └─ if inventory_check_id is null:
        POST /procurement/boms/{id}/inventory-check
POST   /procurement/boms/{id}/export                          → the client PDF
```

30.5 Shortfall to sent PO

```
POST   /procurement/boms/{id}/generate-pos                  → resolved + needs_supplier counts
GET    /procurement/pos?bom_id={id}                          → the list
    for each needs_supplier PO:
GET    /procurement/suppliers?q=...                          → the picker
PATCH /procurement/pos/{po}/items {base_revision, po_edits:{supplier_id}}
    for each PO:
POST   /procurement/pos/{po}/submit                          → pending_approval
POST   /procurement/pos/{po}/approve                          → approved (routed role)
POST   /procurement/pos/{po}/send                            → sent + po_number + PDF
GET    /procurement/pos/{po}                                ⌚ 30-60s → watch the tracking timeline
POST   /procurement/pos/{po}/acknowledge {channel:"call"}    → confirmed (manual path)
```

30.6 Release to production

```
GET    /procurement/production-tracking?order_id={id}        → the board
    when a style reads material_ready:
POST   /procurement/production-tracking/{tracking_id}/transition
        {status:"released_to_production"}                    → Phase 1 takes over
```

31. Quick reference card

The five 409s that gate everything

Gate	Blocks	Unblock with
<code>spec_suggestion_unconfirmed</code>	Generate BOM	<code>POST /order-styles/{id}/attachments</code>
<code>cutting_confirmation_required</code>	Approve BOM	<code>POST /boms/{id}/confirm-cutting</code>
<code>bom_not_approved</code>	Inventory check	<code>POST /boms/{id}/approve</code>
<code>no_inventory_check</code>	Generate POs	<code>POST /boms/{id}/inventory-check</code>
<code>needs_supplier</code>	Submit PO	<code>PATCH /pos/{id}/items</code> with <code>po_edits.supplier_id</code>

The four numbers that flow through everything

BOM	<code>bom_item.bulk_qty</code>	<code>= order_qty × qty_per_garment</code>
CHECK	<code>required_qty</code>	<code>= bom_item.bulk_qty</code>
	<code>shortfall_qty</code>	<code>= required - min(required, available)</code>
PO	<code>po_item.qty</code>	<code>= shortfall_qty</code>
	<code>po_item.amount</code>	<code>= qty × unit_price</code>

The two lock messages

- `stale_revision` → GET, re-present, retry. Never force.
- `po_locked` / `bom_locked` → the record is frozen. Cancel and re-issue.

The three "the system is asking you" moments

Signal	Where	Meaning
<code>spec_match_status: "suggested"</code>	Breakdown	"I found a spec. Is it the right one?"
<code>flags: ["suggestion"]</code>	Inventory check	"This might be the stock item. Confirm?"
<code>candidates.ambiguous: true</code>	PO generation	"Two suppliers are too close to call. You pick."

Each one is a deliberate refusal to guess. Give each a first-class UI affordance — they are where the system's accuracy actually comes from.